CREATION OF CROP YIELD ESTIMATION MODEL USING GEO SPATIAL TECHNOLOGY

A Project report submitted in partial fulfillment of the requirements for the award of a degree of

BACHELOR OF TECHNOLOGY

In

GEO-INFORMATICS ENGINEERING

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DECLARATION

I hereby declare that the work presented in this Project entitled **"CREATION OF CROP YIELD ESTIMATION MODEL USING GEO-SPATIAL TECHNOLOGY"** is an original work done by us under the guidance of Prof. GURRAM MURALI KRISHNA sir, Department of Geo- Engineering & RDT, Andhra University college of Engineering, Visakhapatnam. I hereby declare that no part of the work either partly or fully has been submitted for the award of any Degree or Diploma to any University or Institution.

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BACHELOR OF TECHNOLOGY PROJECT EVALUATION REPORT

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ABSTRACT

Crop yield estimation is one of the most demanding tasks in agriculture. It is crucial in making decisions at global, regional, and field levels. Faultless yield prediction demands elementary understanding of the relationship between yield and the associated factors. Machine learning is an indispensable tool for crop yield prediction, which helps in making decisions with regard to the type of crops to be grown during the season and what to be done during the growing season of crops. An accurate crop prediction model helps farmers in deciding the financial investment and forecast the returns on the investment.

Reliable estimation of region-wide rice yield is vital for food security and agricultural management. Field-scale models have increased our understanding of Crop yield and its estimation under theoretical environmental conditions. Remote Sensing is a useful tool to upscale yield estimates from farm scales to regional levels. Much research used Remote Sensing with rice models for reliable yield estimation.

Crop yield estimation is a critical aspect of modern agriculture. In this project paddy crop is covered. The same method applies to all other crops. With the advent of remote sensing and GIS technologies, it has become possible to estimate crop yields using various methodologies. This study applied space-borne satellite based NDVI to predict crop yield at field level. It was carried out in India, Andhra Pradesh state (East Godavari and West Godavari districts). The purpose of the study was to investigate the relationship between space-borne Satellite based NDVI and rice yield in irrigated fields, and a tool is created in ArcGIS for validation on another study area, in which a single crop NDVI is required as an input and then it automatically generates the crop yield layer as an output based on the NDVI values of the crop area.

Yield estimation using remote sensing is done by using statistical methods, such as regression analysis and modelling in GIS and Excel, including classification and estimation. One popular method for estimating paddy yield is the crop yield estimation model using classified and modelled data. It is also important to validate the developed model on another nearby study area. The identification of crops is an important step in estimating crop yields and managing agricultural resources. They provide accurate and timely information that is critical for managing agricultural resources and increasing crop yields.

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CHAPTER-1
INTRODUCTION

1.1 GENERAL

Agriculture: The science or practice of farming, including cultivation of the soil for the growing of crops and the rearing of animals to provide food, wool, and other products. Agriculture was the key development in the rise of sedentary human civilization, whereby farming of domesticated species created food surpluses that enabled people to live in cities.

Agriculture plays a vital role in the Indian economy. Agriculture is the backbone of our country. It includes farming of crops, animal husbandry, pisciculture, agro-forestry etc. Nearly 60% of the Indian Population primarily depend on agriculture. Agriculture along with fisheries, forestry and other allied sectors contribute around 15.87% to the overall GDP of our country. The particular weather and soil conditions allow for crops in India uniquely suited to it. Let us take a look at the major crops in India.

Crop Seasons in India: In India there are three major crop seasons:

a) Kharif Crop Season

b) Rabi Crop Season

c) Zaid Crop season



a) Kharif season:

- Kharif crops or monsoon crops are sown with the beginning of the first rains in July, during the south west monsoon season in India.
- The Kharif Season lasts from *June* to *October*.
- Kharif means "autumn" in Arabic. Since this period coincides with the beginning of autumn/winter in the Indian subcontinent, it is called the "Kharif period".
- Kharif crops are dependent on the large quantity of rain water as well as its timing.
- Jowar, Rice (Paddy), Millet, Maze (corn), Soyabean, Turmeric, Groundnut, cotton, sugarcane, green gram (moong), sesame (bill), black gram(urad), Tea, Coffee, Rubber are the major crops sown during this season.



Kharif Crops

b) Rabi season:

- The Rabi crops are sown around mid-November, after the monsoon rains are over, and harvesting begins in *April* or *May*.
- The crops are grown either with rainwater that has percolated into the ground, or with irrigation. It also requires cold weather and less water compared to Kharif crops.
- The major rabi crop in India is wheat, followed by barley, mustard, sesame, peas, oats, cereals, pulses, oilseeds, Linseeds etc.



Rabi Crops

c) **Zaid season:**

- In the Indian subcontinent, the crops grown on irrigated lands which do not have to wait for monsoons, in the short duration between Rabi and Kharif crop season, mainly from March to June are called Zaid crops.
- Zaid crops are grown mainly in the summer season during a period called the "Zayad crop season."
- Zaid crops require warm dry weather for major growth periods and longer day length for flowering.
- The main produce is seasonal fruits and vegetables like Watermelon, cucumber, muskmelon, sunflower, sugarcane, Bitter gourd, Pumpkin.



1.2 RICE CROP (PADDY)

Rice is the staple food for more than three billion people worldwide. Rice area accounted for approximately 11.5% of the world's arable land area and it provides almost 19% of the global dietary energy in recent times and its annual average consumption per capita was about 65 kg.

Worldwide, India stands first, after China, in rice area and second in rice production. It contributes 21.5% of the total global rice production (FAO 2000b). Within the country, rice occupies one quarter of the total cropped area, it contributes 40 to 43% of the total food grain and plays an important role in the national food and livelihood security system. A combination of increased area and higher cropping intensity transformed India from a net importer to a potential exporter of quality rice (FAO 2000b).

Therefore, rice area mapping and forecasting its production is vital for food security, where demands often exceed production due to an ever-increasing population. The aim of yield forecast is to give a precise, scientifically sound and independent forecasts of crops' yield as early as possible during the crops' growing season.

1.3 RICE GROWING SEASONS IN INDIA

In India rice is grown under widely varying conditions of altitude and climate. The main rice growing season in the country is the 'Kharif'. It is known as winter rice as per the harvesting time. The sowing time of winter (kharif) rice is June-July and it is harvested in *November-December*.

Winter rice (*kharif rice*) is known as 'Aman' in West Bengal, 'Sali' in Assam, 'Sarrad' in Orissa, 'Agahani' in Bihar and Uttar Pradesh, 'Sarava' in Andhra Pradesh, 'Mundakan' in Kerala and 'Samba/Thaladi' in Tamil Nadu. About 84% of the country's rice crop is grown in this season and generally, medium to long duration varieties are grown in this season.

Summer rice is called as *Rabi rice*. It is known as 'Boro' in Assam and West Bengal, 'Dalua'in Orissa, 'Dalwa' in Andhra Pradesh, 'Punja' in Kerala and 'Navarai' in Tamil Nadu and 'Garma' in Bihar. The sowing time of summer rice is *November* to *February* and harvesting time is *March* to *June*. The area under summer rice is only 9% and early maturing varieties are mostly grown in this season.

1.4 THE NEED FOR CROP YIELD FORECASTING

Forecasting crop yield well before harvest is crucial especially in regions characterised by climatic uncertainties. This enables planners and decision makers to predict how much to import in case of shortfall or optionally, to export in case of surplus. It also enables governments to put in place strategic contingency plans for redistribution of food during times of famine. Therefore, monitoring of crop development and of crop growth, and early yield prediction are generally important.

Crop yield estimation in many countries is based on conventional techniques of data collection for crop and yield estimation based on ground-based field visits and reports. Such reports are often subjective, costly, time consuming and are prone to large errors due to incomplete ground observations, leading to poor crop yield assessment and crop area estimations

Estimating crop yield with satellite based remote sensing has a number of advantages over traditional on ground crop survey methods. Using remote sensing helps to lower costs by avoiding intense labour and time needed for crop surveys. As technology has improved, the cost of satellite data and computing has decreased, making satellite-based crop estimation more appealing. Satellite-based vegetation indices make use of spectral bands, and usually the visible spectrum bands and the near-infrared are of interest for satellite-based crop yield estimation.

The most used vegetation index is the Normalized Difference Vegetation Index (NDVI) due to its simplicity and ability to measure vegetation cover (greenness) quite accurately. However, it may be possible to use improved vegetation indices rather than NDVI, and it may also be possible to find improved functional forms over the linear regression form, for estimating crop yield.

1.5 PROBLEM STATEMENT

Remote sensing has been used extensively as a tool to assess and monitor vegetation parameters, crop vigour and yield estimation. It has been shown that remote sensing data provide systematically high quality spatial and temporal information about land surface features including environmental impacts on crop growth conditions. Most studies have established that there is correlation between NDVI and the green biomass and yield, therefore NDVI can be used to estimate yield before harvesting

However, these studies were mostly done at regional/national level covering very large areas. Since they cover very large areas low resolution images were used resulting in generalisation of the crop condition and yield estimates. The coarse resolution also had a mixture of crops and other non-crop vegetation that contributed to the NDVI value, which later was correlated to the final crop yield

Other studies were done at field level and reported high correlation between NDVI and yield. However, most of these studies were done under research conditions involving very small plots with spectral data being collected with ground-based platforms extended over the plots or low flying platforms. Such conditions enable a large degree of control over many extraneous factors and normally result in high quality data and excellent correlation between the measured and remote sensed data. Quarmby et al showed that NDVI can be used to estimate yield from a test field in Greece and Verma et al (1998), working on gram, found high correlation between

NDVI and dry matter. The potential of regression models to estimate crop yield more accurately under variable management conditions was clearly established.

1.6 OBJECTIVES

Many studies on the relationship between remote sensed data and crop yield have been done either on research involving very small areas with very high degree of control of many parameters affecting crop growth and production or on very large areas, which tend to generalise information. Few studies have applied remote sensing data at farmers' field level to estimate yield. In this study, remote sensed data was used to estimate yield at field level where crop growth and yield is a translation of complex environmental factors including management.

The primary objective is to establish the relationship between remotely sensed NDVI measurements and field level yields by integrating other production factors at field level.

Specific objectives:

1. To establish the relationship between NDVI and field level crop yield in irrigated rice

2. To assess the possibility of using a single date multi-spectral image for yield prediction.

1.7 CONCEPTS

1.7.1 The spectral response of vegetation

Green plants have a unique spectral reflectance influenced by their structure and composition. The proportion of radiation reflected in different parts of the spectrum depends on the state, structure and composition of the plant. In general, healthy plants and dense canopies, will reflect more radiation especially in the near infrared region of the spectrum.

In the visible part of the spectrum (0.4 μm – 0.7 μm), plants absorb light in the blue (0,45 μm) and red (0.6 μm) regions and reflect relatively more in the green portion of the spectrum due to the presence of chlorophyll. High photosynthetic activity will result in lower reflectance in the red region and high reflectance in infrared region of the spectrum. In cases where plants are subjected to moisture stress or other conditions that hinder growth, the chlorophyll production will decrease, this in turn leads to less absorption in the blue and red bands (Dadhwall & Ray 2000, de Wit & Boogaard 2001, Janssen & Huurneman 2001, Woldu 1997).

In the near-infrared portion of the spectrum (0.7 – 2.5 µm), green plants reflectance increases to 40 – 60%. Beyond 1,3 µm, there are dips in the reflectance curve due to absorption by water in the leaves, more free water result in less reflectance.

Figure below shows an ideal reflectance curve from healthy vegetation.

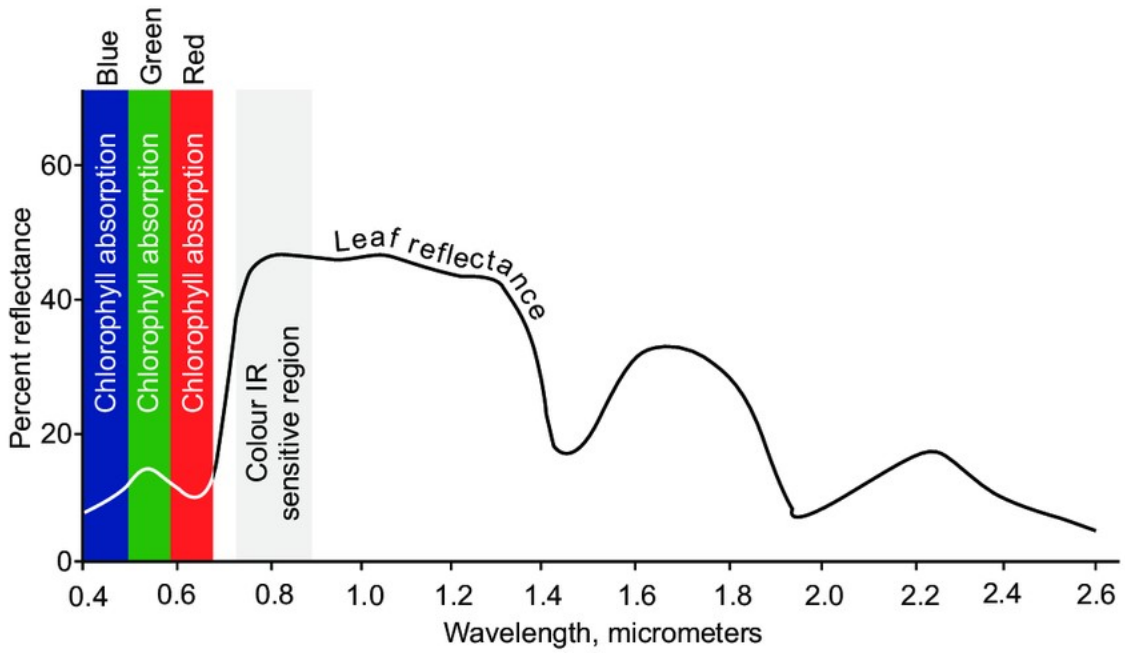


Figure 1.1 Typical Spectral Reflectance Curve for vegetation

As the leaves dry out or as the plant ripens or senescence or become diseased or cells die, there is reduction in chlorophyll pigment. This result in the general increase in reflectance in the visible spectrum and a reduction in reflectance in the middle infrared (MIR) portion of the spectrum due to cell deterioration. Thus, the spectral response of a crop canopy is influenced by the plant health, percentage of ground cover, growth stage, differences in cultural practices, stress condition and the canopy architecture (Verma et al. 1998). This differential reflection of green plants in the visible and infrared parts of radiation makes it possible for the detection of green plants from satellite data because other features on earth surface do not have such unique step-like characteristics in the 0.65 – 0.75 µm spectral range. This signature is unique to green plants only and thus this principle is used in vegetation indices.

1.7.2 Vegetation Indices

Based on this concept, many vegetation indices (VIs) have been developed that have specific features concerning the range of vegetation cover. VIs has been used since the Landsat satellite became operational. The vegetation indices provide information on the state of vegetation on the land surface; vegetation is the result of a complex relation between land and land use, and provides thus a means of monitoring and estimating changes over time (Dadhwall & Ray 2000, de Wit & Boogaard, 2001, Gielen & de Wit 2001).

The commonly used VIs include:

1.7.2.1 Ratio Vegetation Index (RVI)

This is the simplest form of ratio-based vegetation indices calculated through the use of infrared and the red band of the electromagnetic spectrum. It is calculated as follows:

$$RVI = IR/Red$$

where,

RVI = Ratio Vegetation Index

IR = Infrared band of the electromagnetic spectrum.

Red = Red band of the electromagnetic spectrum.

This VI seem to be more affected by the noise present in the image due to atmospheric condition (Gielen & de Wit, 2001).

1.7.2.2 Normalised difference Vegetation Index (NDVI)

This vegetation index is a ratio based VI calculated by the difference of the infrared and red bands as ratio to their sum. Thus:

$$NDVI = \frac{(NIR - Red)}{(NIR + Red)}$$

Where, NDVI = Normalised difference Vegetation Index

NIR = Near Infrared band of the electromagnetic spectrum.

Red = Red band of the electromagnetic spectrum.

Through this normalisation, the values are scaled between –1 and +1.

RVI and NDVI are related to vegetation amount until saturation at full canopy cover and are therefore related to the biophysically active radiation, efficiencies and productivity (Rondeaux et al. 1996).

1.7.2.3 Other Vegetation Indices

Other vegetation indices that take into account the soil effect on vegetation reflectance, especially at low vegetation levels, have been developed. These indices assume a linear relationship between near infrared and the visible reflectance from bare soil. These VIs provide better results than NDVI at low vegetation cover because they eliminate the soil background effect.

These VIs include Perpendicular Vegetation Index (PVI), Weighted Difference Vegetation Index (WDVI), Soil Adjusted Vegetation Index (SAVI), Transformed Soil Adjusted Vegetation Index (TSAVI) and the more recently introduced Modified SAVI (MSAVI).

Out of all VIs, the Normalised Difference Vegetation Index (NDVI) stands out and is regarded as an all-purpose index. This VI is the most widely used and well-understood vegetation index (de Wit & Boogaard 2001). It is simple to calculate, has the best dynamic range and has the best sensitivity to changes in vegetation cover (Gielen & de Wit 2001). It has been found to correlate better with yield than other vegetation indices and thus continues to be used as a vegetation/biomass indicator using remotely sensed data.

1.7.3 Rice growth and production - Phenological stages

The rice life cycle is mainly divided into three phases:

1. **Vegetative phase**: from germination to panicle initiation, characterised by active tillering, gradual increase in plant height and leaf emergency,
2. **Reproductive phase**: from panicle initiation to flowering phase.
3. **Ripening phase**: from flowering to grain maturity.

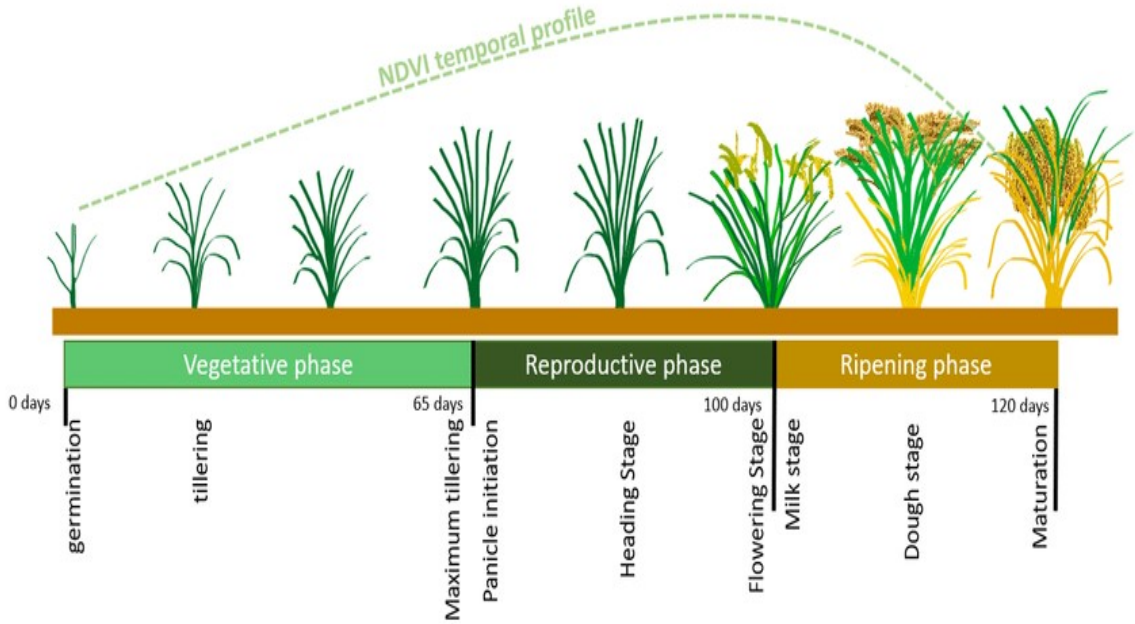
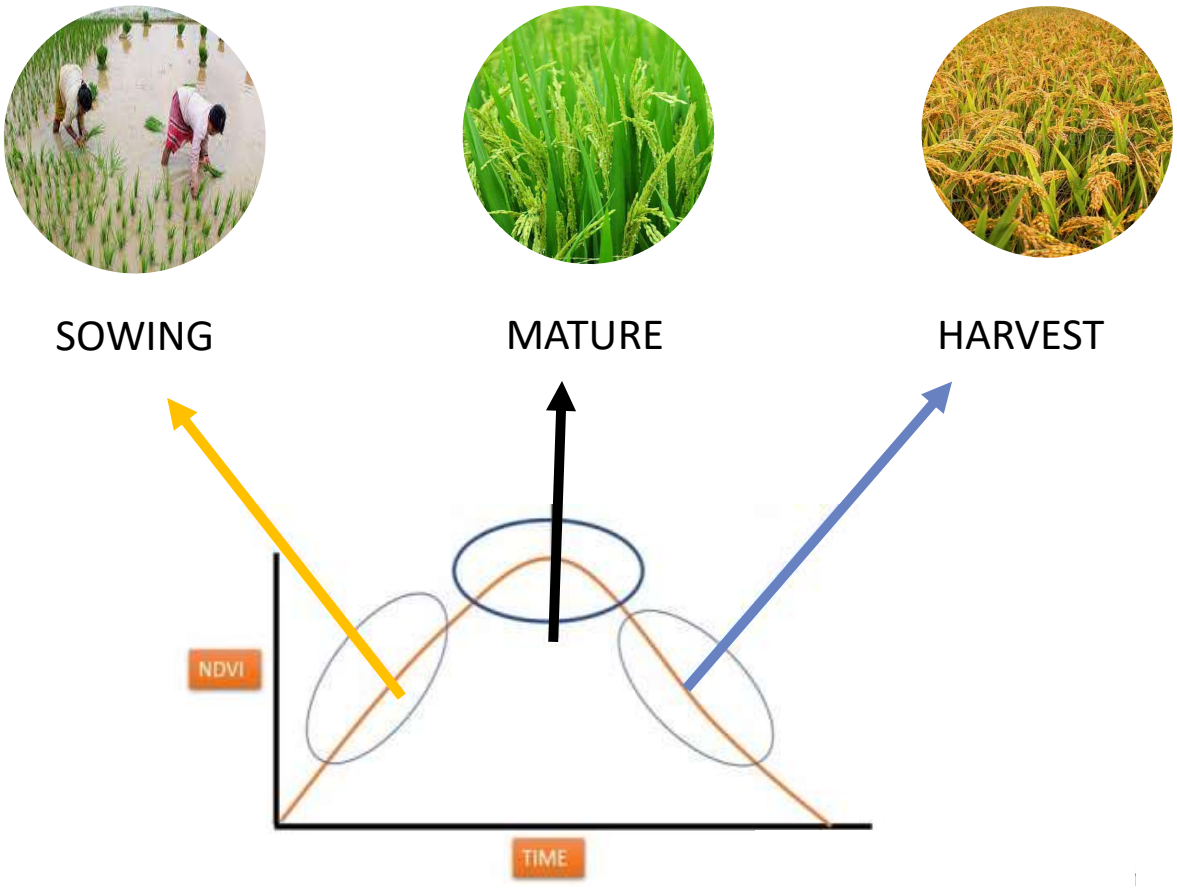


Figure 1.2 Phenological stages of Rice Crop and NDVI Temporal Profile



1.7.4 Hypotheses

Vegetation density is the most obvious physical representation of subsequent yield from crops. The density and health can be monitored using remotely sensed images that measure chlorophyll activity and vegetation vigour. The spectral reflectance is a manifestation of all important factors affecting the agricultural crop and cumulative environmental impacts on crop growth therefore remote sensed data could be used to monitor crop condition through NDVI

Management practices in the production system and how land is utilised will have an effect on the overall productivity. In this aspect, crop growth and crop yield is a response to the type of management and the quality of the land unit

Based on the above, hypothesis adopted in this study are as follows

1. There is significant relationship between NDVI, and yield at field level as:

Crop yield is a *function of NDVI* (Normalized Difference Vegetation Index)

$$YIELD = f(NDVI)$$

$$YIELD = a.e^{(b.NDVI)}$$

where,

a and *b* are coefficients that are determined by fitting the model to a set of data points.

2. There is significant relationship between NDVI, field level management and land.

$$NDVI = f(LAND, MANAGEMENT)$$

3. Single date multi-spectral images can be used to predict yield at field level.

Integrating hypothesis 1 and 2 leads to:

$$YIELD = f(NDVI, LAND, MANAGEMENT)$$

Where only those land and management factors remain relevant that have no impact/relation with NDVI, the rest are now no longer relevant.

CHAPTER-2
LITERATURE REVIEW

Literature review is a critical analysis of existing literature and research on a particular topic that is relevant to the thesis. It involves a comprehensive examination of all the relevant literature and research that has been published on the topic, including academic journals, books, and other credible sources. The purpose is to identify gaps in knowledge or understanding, highlight key findings and concepts, and provide a framework for the research to be conducted in the thesis.

Shania Miralda (2022) studied "*Satellite Based Remote Sensing for Estimating Crop Yield: Examining the Use of Various Functional Forms and Vegetation Indices."*

This study compared various satellite-based vegetation indices for estimating crop yield. NDVI is the most commonly used vegetation index for agricultural or ecology purposes. However, there are a few limitations of NDVI such as saturation with high density biomass, such as high crop yield or very green crops nearing maturity. Therefore, this paper discusses nine other vegetation indices to understand their performance compared to NDVI for estimating crop yield. For corn, soybeans, spring wheat, and winter wheat, the top three indices that provide highest adjusted R^2 value are GOSAVI, GSAVI, and RDVI. Interestingly, OSAVI and SAVI indices, have same equation as GOSAVI and GSAVI except they used red band instead of green band. However, they are not performing any better than NDVI for estimating crop yield.

Mahamadou Belem et al. (2017) published a research article on "*Using NDVI for Crop Yield Prediction in the Sahel Region"*. The Sahel region is a semi-arid zone that is prone to droughts and has a high risk of food insecurity. Accurate predictions of crop yields are crucial for food security planning and decision-making. Remote sensing techniques, such as NDVI, can provide valuable information on crop growth and productivity. NDVI is calculated using the difference in reflectance between the near-infrared and red bands of the electromagnetic spectrum. It is commonly used as an indicator of vegetation greenness and density. Higher NDVI values typically indicate healthier vegetation and higher crop yields. The study used satellite imagery and ground-truthed data from the National Agricultural Research Institute of Niger to develop a crop yield prediction model for the Sahel region. The model was based on multiple linear regression and used NDVI as a predictor variable. The results of the study showed that the NDVI-based model was able to accurately predict crop yields in the Sahel region, with an overall accuracy of 77%. The

model was also able to identify areas of low and high productivity, which can be useful for targeting agricultural interventions and improving food security in the region.

Binbin He et al. (2019) has done a research on "*Crop yield forecasting using remotely sensed vegetation indices and machine learning techniques"*. The study uses data from the US Department of Agriculture (USDA) National Agricultural Statistics Service (NASS) to develop a model for predicting corn and soybean yields in the United States. The model is based on multiple linear regression and uses vegetation indices, including the Normalized Difference Vegetation Index (NDVI), as predictor variables. The results of the study show that the NDVI-based model has high accuracy for predicting corn and soybean yields in the United States, with an overall accuracy of 82% and 75%, respectively. The study also compares the NDVI-based model with other machine learning algorithms, including Random Forest and Artificial Neural Networks, and finds that the NDVI-based model outperforms the other algorithms in terms of accuracy and simplicity. The study demonstrates the potential of using remotely sensed vegetation indices and machine learning techniques for crop yield forecasting, which can provide valuable information for decision-making in agriculture and food security planning. The results of the study can also help inform policies and interventions aimed at improving crop yields and food security in the United States and other regions.

Xiaowei Yu et al. (2018) has published an article on *"Assessment of NDVI-Based Vegetation Indices for Estimating Rice Yield in Northeast China".* The study uses remote sensing data and ground-truthed yield data from six counties in Northeast China to develop and evaluate the performance of various NDVI-based vegetation indices. The study compares different vegetation indices, including the Normalized Difference Vegetation Index (NDVI), Enhanced Vegetation Index (EVI), Soil-Adjusted Vegetation Index (SAVI), and Difference Vegetation Index (DVI), in predicting rice yield. The results of the study show that the NDVI-based vegetation indices are effective for estimating rice yield in Northeast China, with the NDVI having the highest correlation coefficient with rice yield. The study demonstrates the potential of using NDVI-based vegetation indices for estimating rice yield in Northeast China, which can provide valuable information for decision-making in agriculture and food security planning. The results of the study can also help inform policies and interventions aimed at improving rice yield and food security in the region.

Jingjing Liu et al. (2019) published an article on *"Crop classification using machine learning algorithms and multi-temporal Sentinel-2 data"*. The study uses Sentinel-2 satellite imagery and

ground-truthed data from a farm in Hebei Province, China, to develop and evaluate the performance of various machine learning algorithms for crop classification. The study compares different algorithms, including Random Forest, Support Vector Machine, and Deep Learning, in classifying six major crops in the study area, including wheat, corn, soybean, cotton, peanut, and sweet potato. The results of the study show that machine learning algorithms, particularly Random Forest and Support Vector Machine, are effective for crop classification.

Md. Bodruddoza Mia et al. (2019) published an article on *"A review of remote sensing-based crop classification."* The review covers various aspects of remote sensing-based crop classification, including data sources, pre-processing techniques, feature extraction methods, classification algorithms, and accuracy assessment. The article discusses the advantages and limitations of different methods and techniques for crop classification and provides insights into the future directions of research in this field. The review identifies several challenges and issues related to remote sensing-based crop classification, including the complexity of crop spectral signatures, the effects of environmental factors on crop growth and development, and the need for accurate ground-truth data for validation and calibration of classification results. The review highlights the importance of remote sensing-based crop classification for agricultural monitoring and management, crop yield estimation, and food security planning.

R. K. Nayak et al. (2018) made his research on *"Modelling the yield of winter wheat in response to climatic and soil factors using an exponential equation".* The study used data from field experiments conducted in India to develop and evaluate the performance of the yield model. The study included data on winter wheat yield, weather variables such as temperature, rainfall, and solar radiation, and soil characteristics such as soil moisture and pH. The yield model developed in the study is based on the principle that the yield of winter wheat is primarily determined by the interaction of weather variables and soil characteristics. The model incorporates an exponential equation to describe the relationship between winter wheat yield and weather and soil variables. The results of the study show that the yield model developed in the study provides a good estimate of winter wheat yield, with high accuracy and precision. The study also finds that the model is sensitive to changes in weather and soil variables, and can be used to predict winter wheat yield under different environmental conditions. The equation had a high correlation with observed yields, with an R^2 value of 0.92.

L. M. Cangar et al. (2021) has done a research on “Development and validation of a crop yield model for maize using exponential and polynomial equations". The study developed and validated a crop yield model for maize using exponential and polynomial equations. The model incorporated climate variables, soil properties, and management practices and had an R2 value of 0.83. The study uses data from field experiments conducted in the Philippines to develop and evaluate the performance of the yield model. The study includes data on maize yield, weather variables such as temperature, rainfall, and solar radiation, and soil characteristics such as soil moisture and pH. The yield model developed in the study is based on the principle that maize yield is primarily determined by the interaction of weather variables and soil characteristics. The model incorporates exponential and polynomial equations to describe the relationships between maize yield and weather and soil variables. The results of the study show that the yield model developed in the study provides a good estimate of maize yield, with high accuracy and precision. The study also finds that the model is sensitive to changes in weather and soil variables, and can be used to predict maize yield under different environmental conditions.

Conclusion:

In conclusion, The Normalized Difference Vegetation Index (NDVI) is a widely used remote sensing index for monitoring crop growth and yield estimation. NDVI is calculated using the difference between the red and near-infrared bands of the electromagnetic spectrum, and its value typically ranges from -1 to 1, with higher values indicating denser vegetation. The NDVI index is a powerful tool for estimating crop yields, and its performance can be further enhanced by using machine learning algorithms and selecting appropriate vegetation indices. And exponential equations are commonly used in crop yield modelling because they can capture the nonlinear relationship between yield and its predictors. Crop yield equations based on exponential equations have been developed and validated for various crops and can incorporate climate variables, soil properties, and management practices.

CHAPTER – 3

STUDY AREA DESCRIPTION

3.1 AREA CONSIDERED FOR MODEL CREATION

- The area selected for present study is East Godavari, Andhra Pradesh. (14266.849 Km^2)
- The coordinates of Study Area are (17°19’ 16.5" N, 82°02’16.5" E) under WGS 1984 UTM Zone 44N.

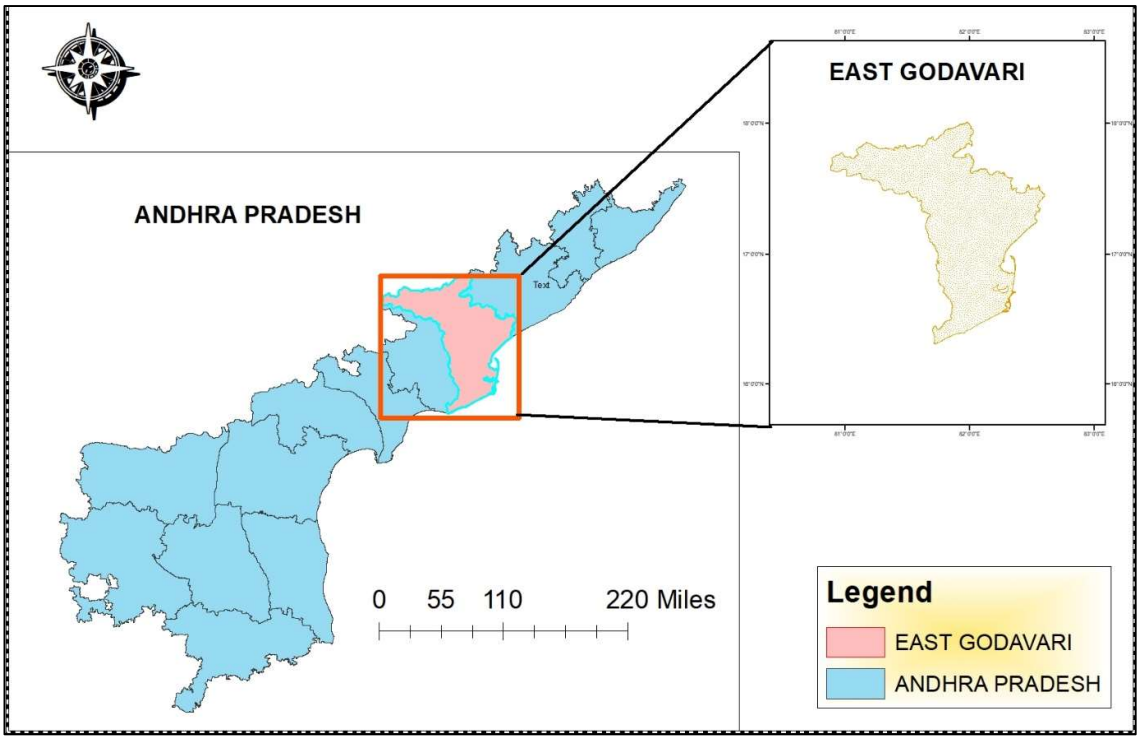


Figure 3.1 Area Considered for Model development

Geography of the East Godavari district, as the name suggests, is closely associated with the Godavari River which occupies a major portion of the delta area. East Godavari district in Andhra Pradesh, India, has a tropical climate, with hot and humid summers and moderate winters. The district receives most of its rainfall during the monsoon season, which lasts from June to September.

The average temperature in the district ranges from 25°C to 35°C during the summer months from March to June. The humidity levels during this time are also high, making the weather quite uncomfortable. The district experiences occasional heatwaves during this period,

with temperatures reaching as high as 45°C.The monsoon season in East Godavari brings moderate to heavy rainfall, with the district receiving an average of around 1,000 mm of rainfall annually. The monsoon season lasts from June to September, with July and August being the wettest months.

The winter season in the district, from December to February, is mild and pleasant, with temperatures ranging from 18°C to 25°C. The weather during this time is generally dry and comfortable, making it an ideal time to visit the region. Overall, the climate of East Godavari district is characterized by hot and humid summers, a wet monsoon season, and mild winters, typical of a tropical climate.

3.1.1 East Godavari soil types

East Godavari district in Andhra Pradesh, India, has diverse soil types due to its topography, geology, and climatic conditions. The district has six major soil types, which are as follows:

a) **Red Sandy Loam Soil:** This soil type is found in the upland areas of the district, which are predominantly hilly and undulating. The soil is reddish-brown in color and has low fertility, but it is suitable for growing crops such as cashew, coconut, and oilseeds.
b) **Red Loamy Soil:** This soil type is found in the midland areas of the district, which are moderately undulating. The soil is reddish-brown in color and has good fertility, making it suitable for growing crops such as paddy, sugarcane, and pulses.
c) **Alluvial Soil:** This soil type is found in the low-lying areas of the district, which are predominantly flat and have a high-water table. The soil is rich in organic matter and minerals, making it highly fertile and suitable for growing crops such as paddy, sugarcane, and vegetables.
d) **Black Cotton Soil:** This soil type is found in the northern parts of the district, which are predominantly flat and have low rainfall. The soil is black in color and has high clay content, making it difficult to cultivate. However, it is suitable for growing crops such as cotton, groundnut, and red gram.
e) **Laterite Soil:** This soil type is found in the eastern parts of the district, which are predominantly hilly and undulating. The soil is red in color and has low fertility, but it is suitable for growing crops such as cashew and coconut.

f) **Coastal Sandy Soil:** This soil type is found in the coastal areas of the district, which have high saline content due to their proximity to the sea. The soil has low fertility and is suitable for growing crops such as coconut and cashew.

Overall, the soil types in East Godavari district are diverse and offer ample opportunities for agriculture and horticulture. Farmers in the district use different farming techniques and practices depending on the soil type and crop grown

3.1.2 East Godavari Crop Production

East Godavari district in Andhra Pradesh, India, is known for its diverse agricultural production, with a variety of crops being grown throughout the district. The district's fertile soil, favourable climate, and irrigation facilities have made it a major agricultural hub in the region. Here are some of the major crops grown in the district:

a) **Paddy**: Paddy is the main crop grown in the district, especially in the delta regions. The district is one of the major producers of paddy in Andhra Pradesh, with an average yield of around 2,500 kg per hectare.
b) **Sugarcane**: Sugarcane is another major crop grown in the district, especially in the low-lying areas. The district has several sugar factories, and sugarcane is also used for jaggery production.
c) **Coconut**: Coconut is a major horticultural crop in the district, especially in the coastal areas. The district is one of the largest producers of coconut in Andhra Pradesh, with an average yield of around 10,000 nuts per hectare.
d) **Cashew**: Cashew is another major horticultural crop grown in the district, especially in the upland areas. The district is one of the major producers of cashew nuts in Andhra Pradesh, with an average yield of around 700 kg per hectare.
e) **Vegetables**: A variety of vegetables such as tomato, brinjal, chilli, and okra are grown in the district, especially in the midland areas. The district has several vegetable markets, and vegetables are also exported to other states.
f) **Oilseeds**: Oilseeds such as groundnut, sesame, and sunflower are also grown in the district, especially in the upland areas. The district has several oil mills, and oilseeds are also exported to other states.

3.2 AREA CONSIDERED FOR MODEL VALIDATION

- The area selected for present study is *West Godavari*, Andhra Pradesh, India (2178 km²).
- The coordinates of Study Area are (16°55’ 2.7" N, 81°20’23.79"E)

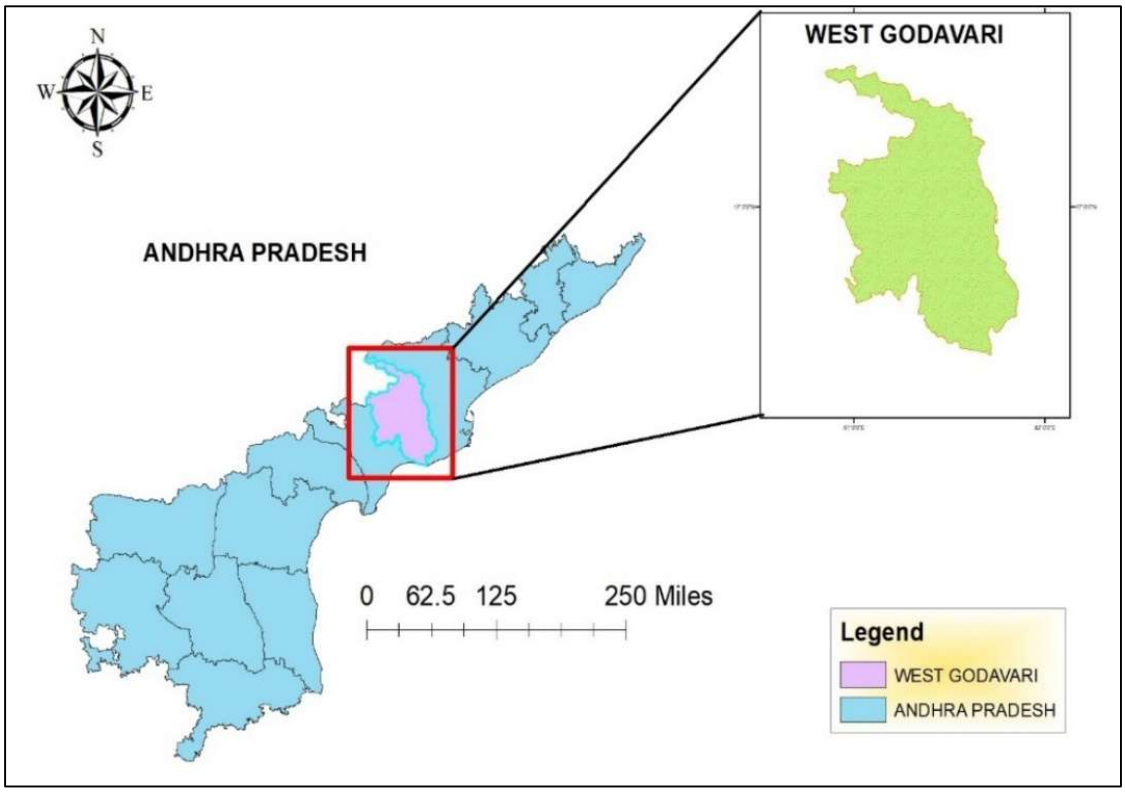


Figure 3.2 Area considered for Validation

3.3 SOFTWARE USED

- **ArcMap 10.8:**

  ArcMap 10.8 is a geospatial software to view, edit, manage and analyze geographic data. Esri developed ArcMap 10.8 for mapping on desktop, mobile, and web. It allows you to create your own maps from scratch starting with geographic data and to analyze data that has a locational component. ArcMap 10.8 uses the concept of a Geographic Information System (GIS) to build maps in which each category of spatial feature is a separate layer. In this study, we have used various data management tools for clipping, resampling and spatial analyst tools for extraction and raster calculator and for creating layouts provided by this software.

- **Google Earth Pro:**

Google Earth Pro is a free geospatial desktop application that allows you to see the world and create highly detailed maps. Aimed at users with advanced feature needs, Earth Pro's 3D mapping system allows you to import and export GIS data, go back in time with historical imagery, and analyze and capture geographical data. Users can explore the globe by entering addresses and coordinates, or by using a keyboard or mouse. In addition to Earth navigation, Google Earth Pro provides a series of other tools through the desktop application, including a measure distance tool and draw polygon. Here we used Google Earth Pro to improve the accuracy of crop classification.

- **Microsoft Excel:**

Microsoft Excel has the basic features of all spreadsheets, using a grid of cells arranged in numbered rows and letter-named columns to organize data manipulations like arithmetic operations and functions. In this project Excel sheets are used for Statistical operations and to plot bar graphs.







CHAPTER - 4

METHODOLOGY

The basic goal of this study is to create a tool (model) in ArcGIS which generates crop yield map by using crop NDVI of study area as an input. The Methodology is explained in detail with a flow chart below:

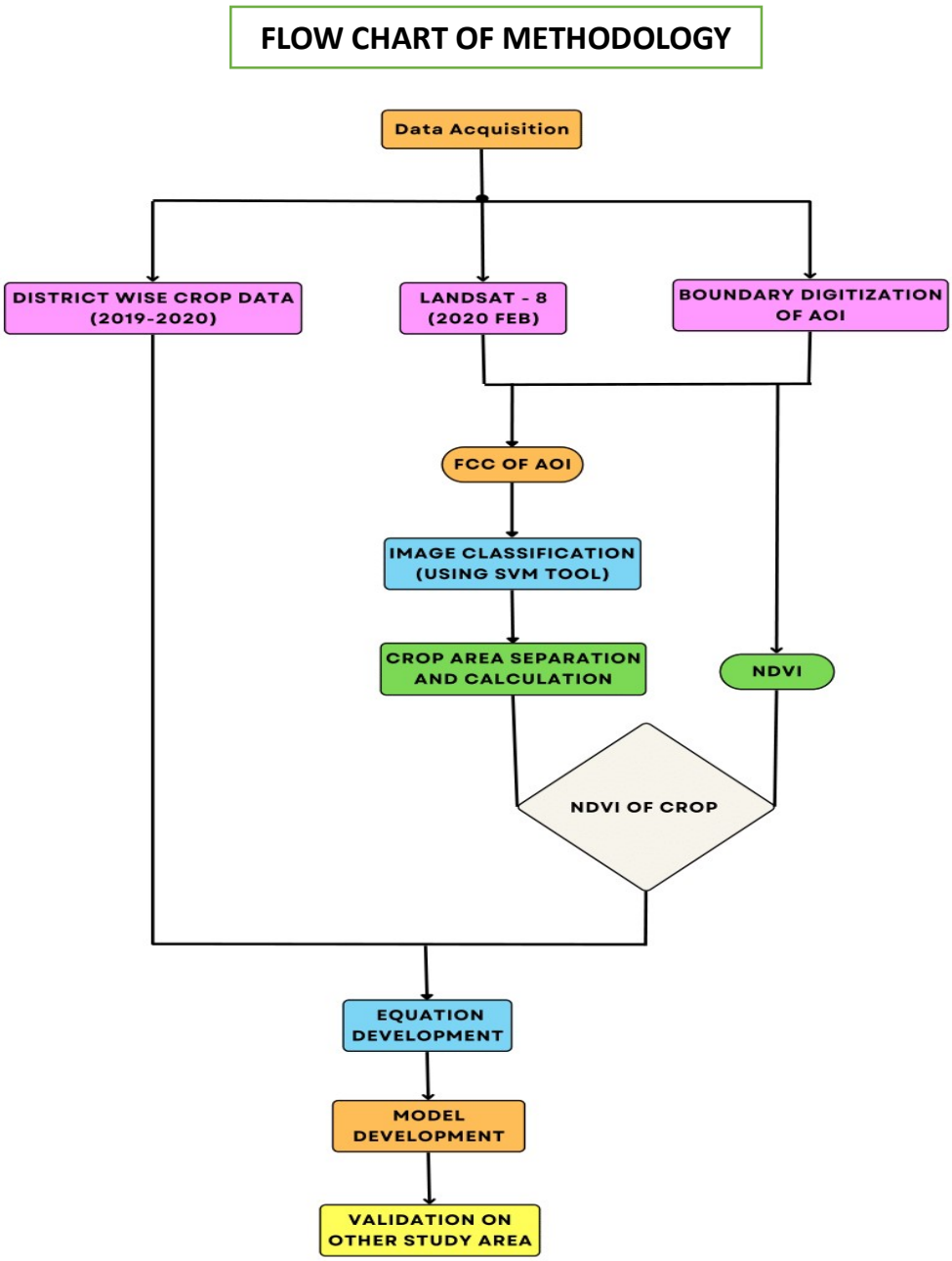


Figure 4.1 Flow Chart of Methodology

4.1 DATA COLLECTION:

Primary data we need is the crop data that can be collected from agricultural websites and other online sources.

4.1.1 Download crop data

1) *Crop Production Statistics Information System* provides the statistical data related to agriculture production and yield of every state of India up to district level. This Information System is under the Ministry of Agriculture and Farmers Welfare, Government of India. To access district-level data, you can start by visiting their website at https://aps.dac.gov.in and exploring their data.
2) ICRISAT (*International Crops Research Institute for the Semi-Arid Tropics*) is a non-profit agricultural research organization that collects and provides various types of agricultural data, including district-level data, for different regions around the world. However, the availability of district-level data may vary depending on the region and the type of data being requested. To access ICRISAT district-level data, you can start by visiting their website at www.icrisat.org and exploring their data and tools section.

Table 4.1 District wise Rice Crop Statistics of Andhra Pradesh in 2019-2020 Rabi Season

Crop Production Statistics

State/Crop/District	Year	Season	Area (Hectare)	Production (Tonnes)	Yield (Tonnes/Hectare)
Andhra Pradesh					
Rice					
1.ANANTAPUR	2019-20	Rabi	14331	44922	3.13
2.CHITTOOR	2019-20	Rabi	43855	157410	3.59
3.EAST GODAVARI	2019-20	Rabi	166544	888271	5.33
4.GUNTUR	2019-20	Rabi	26430	105191	3.98
5.KADAPA	2019-20	Rabi	9162	29795	3.25
6.KRISHNA	2019-20	Rabi	71173	315771	4.44
7.KURNOOL	2019-20	Rabi	38714	151604	3.92
8.PRAKASAM	2019-20	Rabi	50503	191440	3.79
9.SPSR NELLORE	2019-20	Rabi	215725	925460	4.29
10.SRIKAKULAM	2019-20	Rabi	9520	30457	3.20
11.VISAKHAPATANAM	2019-20	Rabi	6559	18873	2.88
12.VIZIANAGARAM	2019-20	Rabi	7266	22985	3.16
13.WEST GODAVARI	2019-20	Rabi	169775	915879	5.39
Total - Rice			829557.00	3798058	4.58

Source: https://aps.dac.gov.in/APY/Public_Report1.aspx

4.1.2 Download Satellite Data:

Based on the Crop data, download the satellite imagery of the study area in the time period where the crop is in mature stage i.e., For Rabi season in 2019-2020, The sowing period of rice crop is in December or January and the harvesting period is in March or April. Hence the rice crop attains maturity in the month of February or March. So, download the satellite imagery in the month of February or March in the year 2020.

Satellite Imagery	Year	Acquired Date	Resolution	Sensors
LANDSAT 8	2020	23/02/2020	30 m	ETM+, OLI

Satellite images used for this study include Landsat-8 image of East Godavari taken on 23rd February 2020 with a spatial resolution of 30m.

Table 4.2 Band details of Landsat 8 data product

Landsat 8 Bands	Wavelength (µm)	Resolution (meters)
Band 1 - Coastal aerosol	0.43-0.45	30
Band 2 - Blue	0.45-0.51	30
Band 3 - Green	0.53-0.59	30
Band 4 - Red	0.64-0.67	30
Band 5 - Near Infrared (NIR)	0.85-0.88	30
Band 6 - Shortwave Infrared (SWIR) 1	1.57-1.65	30
Band 7 - Shortwave Infrared (SWIR) 2	2.11-2.29	30
Band 8 - Panchromatic	0.50-0.68	15
Band 9 - Cirrus	1.36-1.38	30
Band 10 - Thermal Infrared (TIRS) 1	10.6-11.19	100
Band 11 - Thermal Infrared (TIRS) 2	11.50-12.51	100

4.1.3 Download AOI Shapefile:

To download India administrative boundary shapefiles from igismap, follow these steps:

1. Go to the igismap website at https://igismap.com/.
2. Click on the "Download" tab at the top of the page.
3. Scroll down to the "India Shapefile" section and click on the "Download Now" button.
4. You will be directed to a page where you can select the administrative level you want to download, such as states, districts, or blocks.
5. Select the administrative level you want and click on the "Download" button next to it.
6. A zip file containing the shapefiles for the selected administrative level will be downloaded to your computer.

4.2 FCC GENERATION OF STUDY AREA

FCC (*False Color Composite*) is useful for analyzing vegetation. In particular, areas in red have better vegetation health. Dark areas are water and urban areas are white. It uses near-infrared (Band 5), red (Band 4), and green (Band 3). Because chlorophyll reflects near-infrared light.

Tool Used: ***composite Bands***

Data Management Tools ⟶ Raster ⟶ Raster Processing ⟶ **Composite Bands**

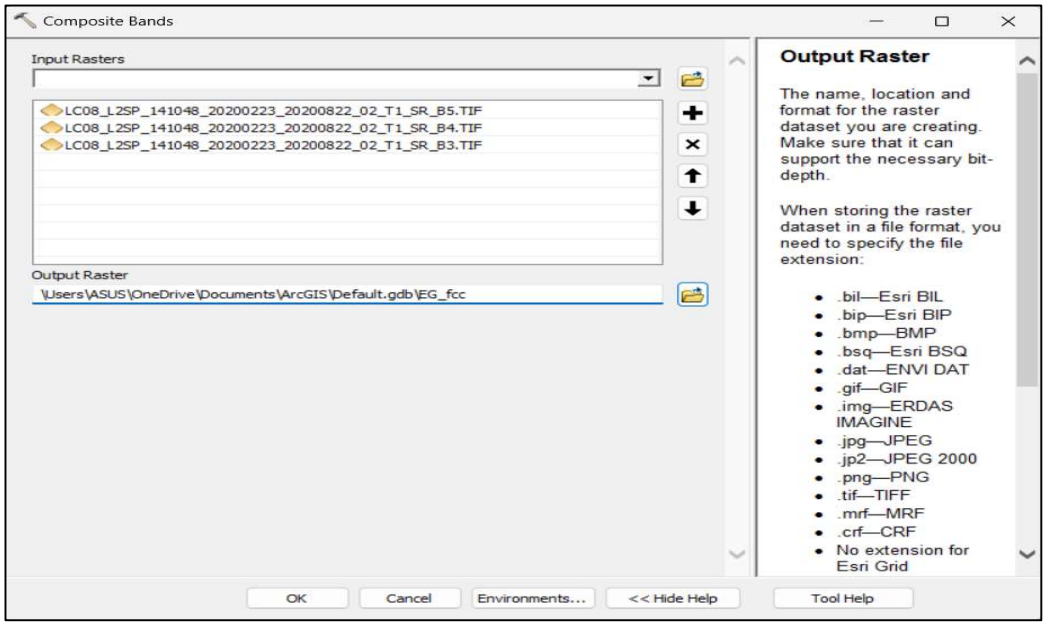


Figure 4.2 Input for FCC generation

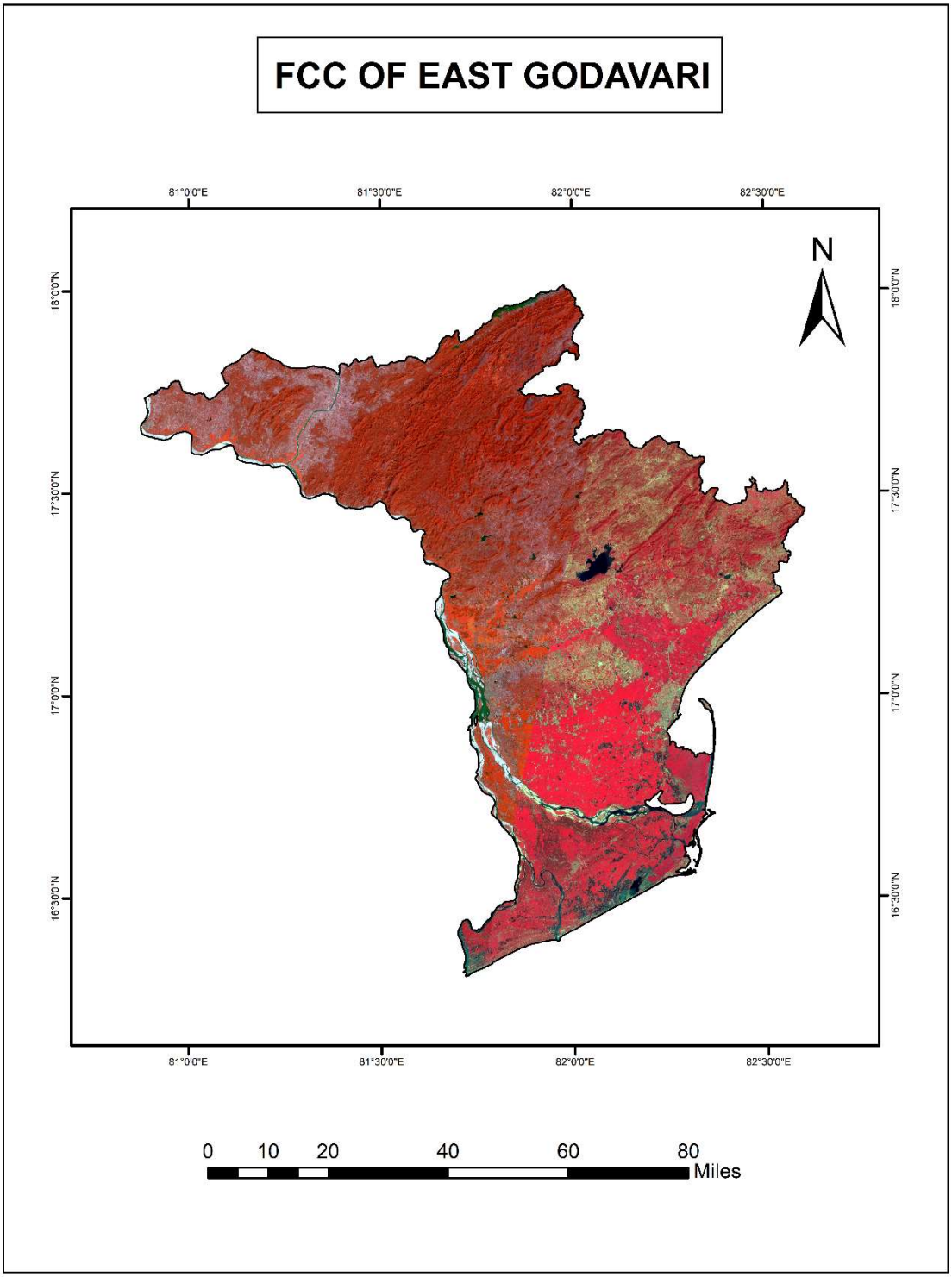


Figure 4.3 Output: FCC of East Godavari

4.3 CROP CLASSIFICATION

4.3.1 Crop Sampling

In order to classify or extract the rice crop, we have to create training samples of each class.

The following are the six classes considered for Classification:

Table 4.3 Classes and their Color Signature in FCC

Class	Color in FCC
Rice Crop	Bright Red
Vegetation and Other Crops	Dark Red
Fallow Land and Barren Land	Pinkish yellow
Built-up	Light blue or cyan
Waterbodies	Black or Dark blue
Sandbar	Bright White

To create the *training sample file*, use the Training Sample Manager from the *Image Classification toolbar.*

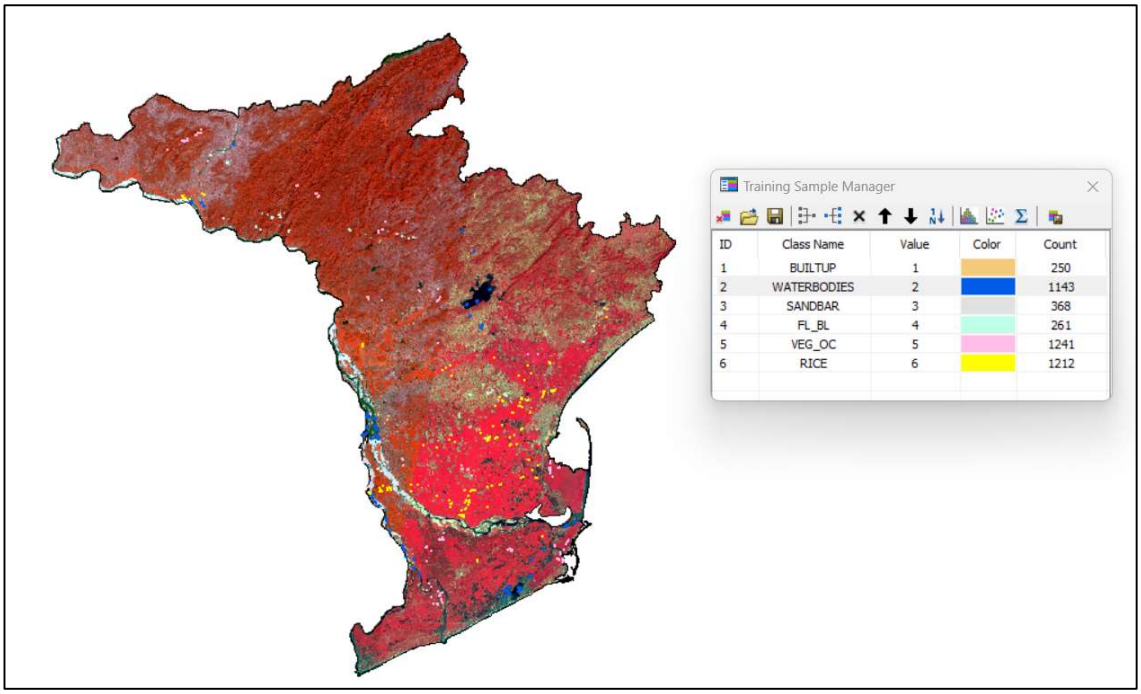


Figure 4.4 Training Samples with classes

4.3.2 Crop classification using ML tool in ArcGIS

Support Vector Machine (SVM): The SVM tool in ArcGIS is a machine learning algorithm that is used for classification and regression analysis. SVMs are particularly useful for analyzing spatial data and are often used for land cover classification, crop classification, and urban land use mapping.

The SVM classifier is a powerful supervised classification method. It is well suited for segmented raster input but can also handle standard imagery. It is a classification method commonly used in the research community. For standard image inputs, the tool accepts multiple-band imagery with any bit depth, and it will perform the SVM classification on a pixel basis, based on the input training feature file.

Tool Used: ***Train Support Vector Machine Classifier***

It generates an Esri classifier definition (.ecd) file using the SVM classification definition.

Spatial Analyst Tools → Segmentation and Classification → **Train SVM Classifier**

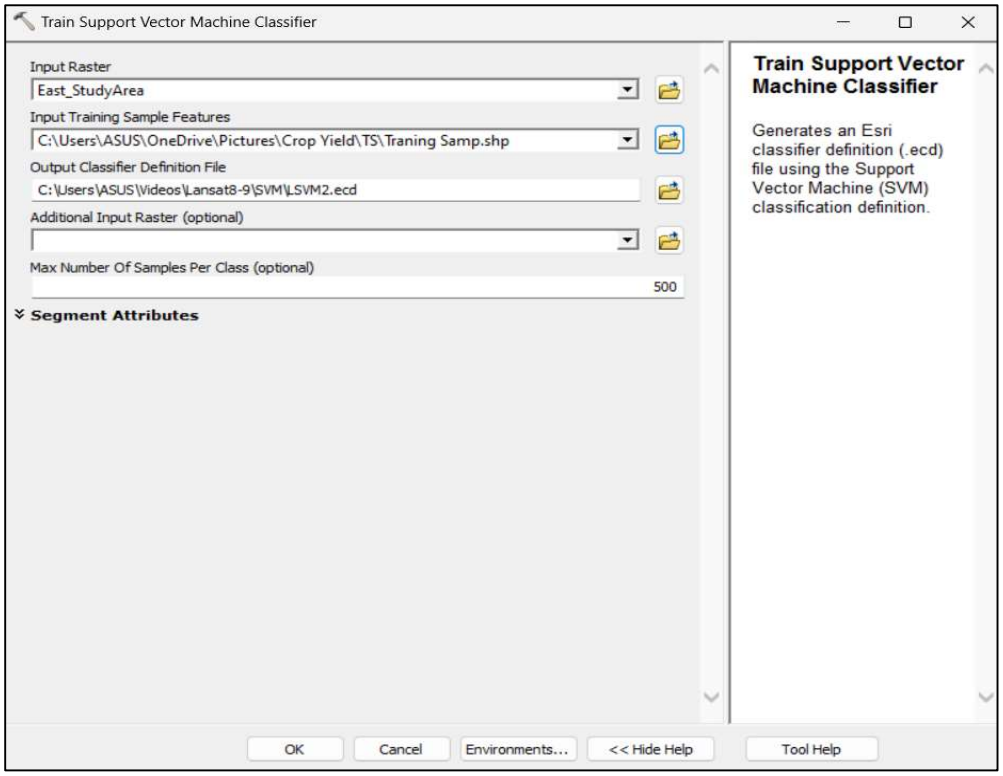


Figure 4.5 Input for SVM Classifier

OUTPUT: Esri classifier definition (.ecd) file is generated with the name "LSVM2.ecd".

Tool Used: ***Classify Raster***

It classifies a raster dataset based on an Esri classifier definition (.ecd) file and raster dataset inputs. The (.ecd) file contains all the information needed to perform a specific type of Esri-supported classification.

Spatial Analyst Tools ⟶ Segmentation and Classification ⟶ Classify Raster

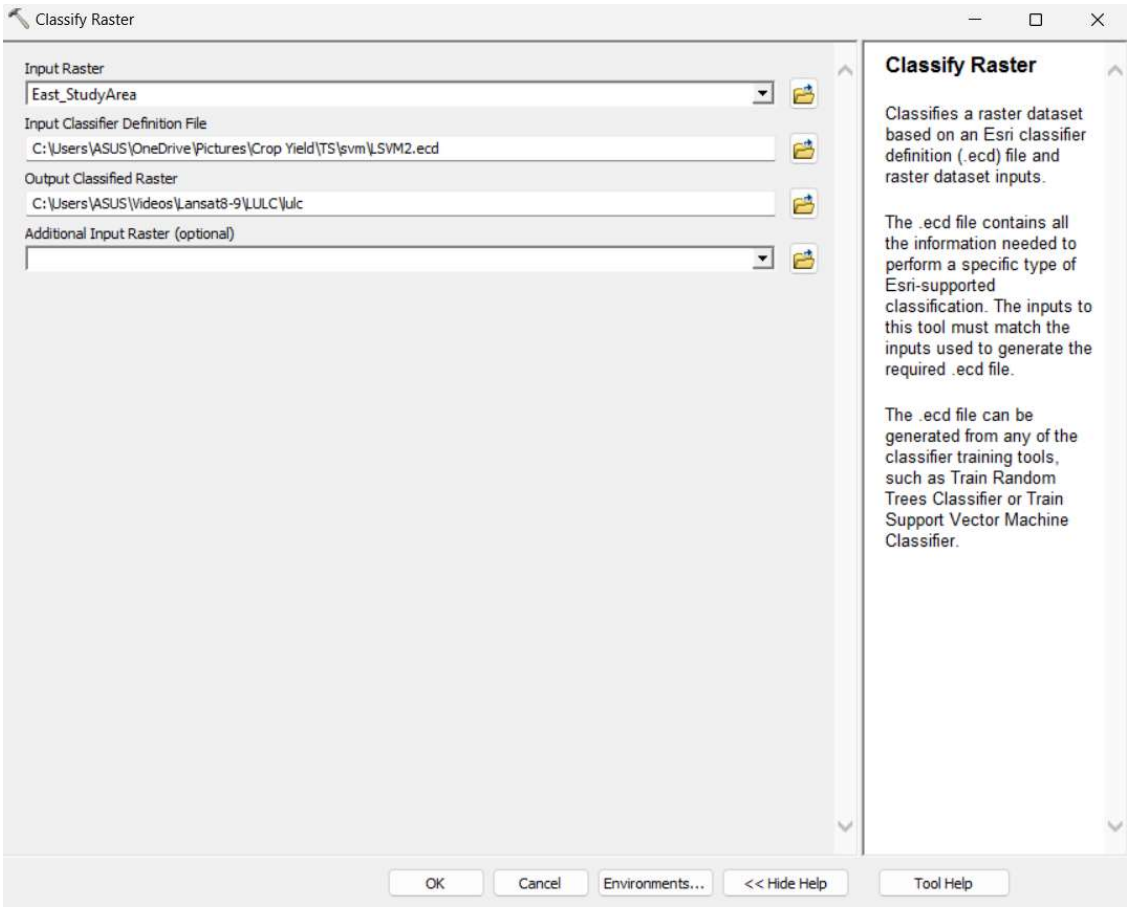


Figure 4.6 Input for Classify Raster

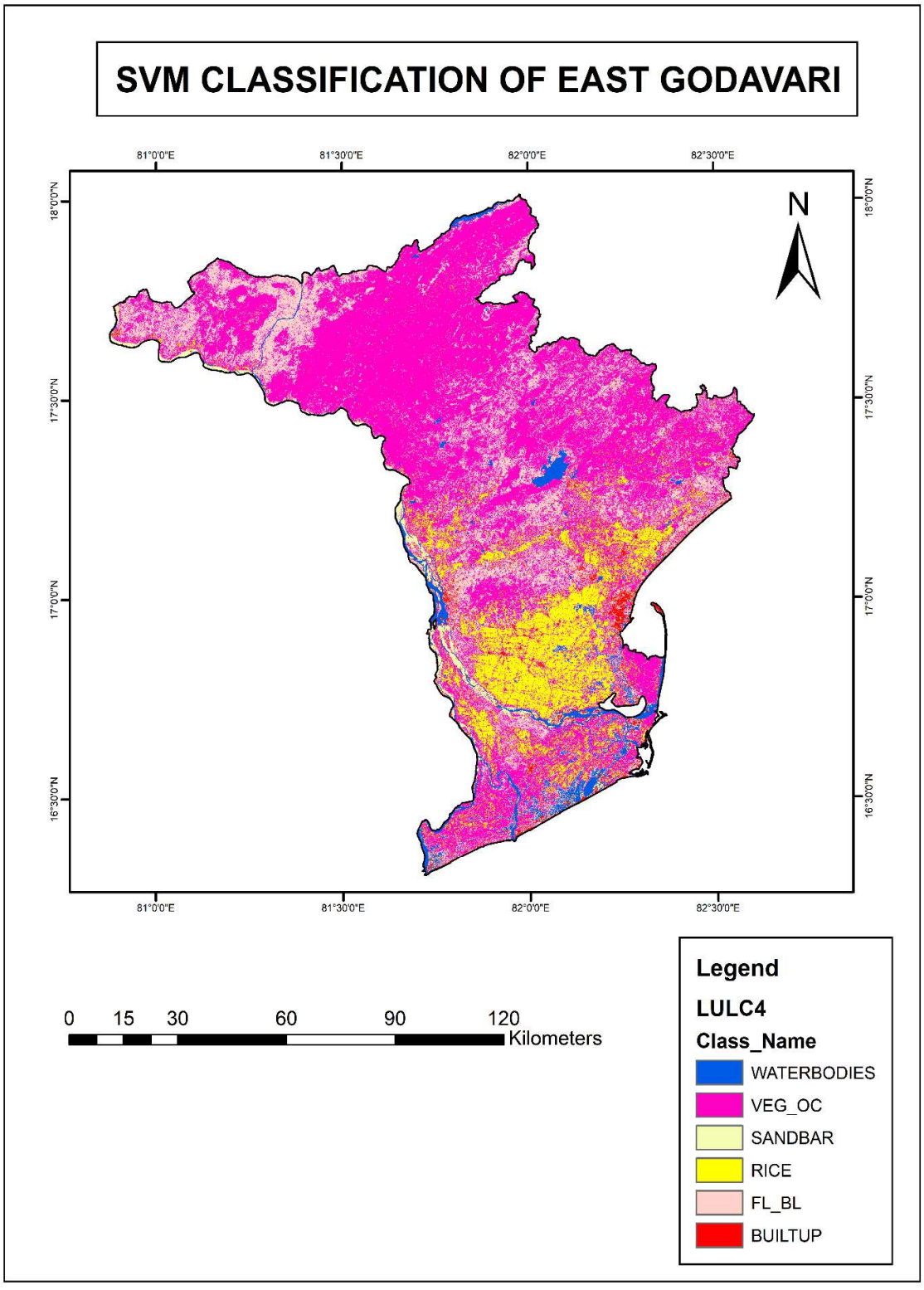


Figure 4.7 output: SVM Classification map of Study Area

4.4 CROP AREA CALCULATION

In this step, we are going to calculate the Rice Crop area obtained in the SVM classified map. For this we have to calculate the number of pixels indicating the class rice crop in the SVM classified map.

Tool Used: ***Build Raster Attribute Table***

It creates or update a table with information about the classes in your raster datasets. This is used primarily with discrete data. It generates the count of pixels in each class.

Data Management Tools ⟶ Raster ⟶ Raster Properties ⟶ **Build Raster Attribute**

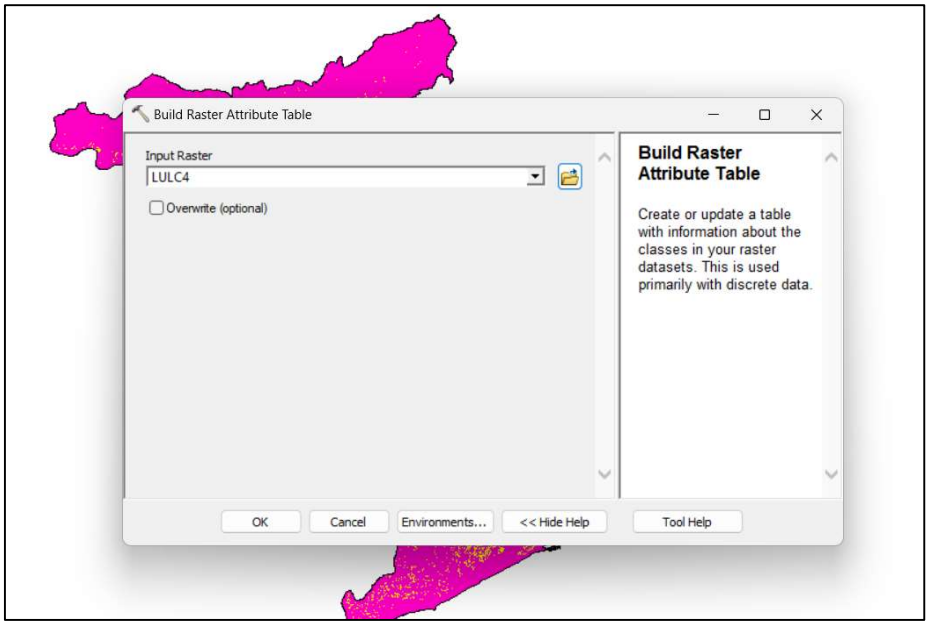


Figure 4.8 Input for Build Raster Attribute Table

Table

LULC4

Rowid	VALUE	CLASSVALUE	CLASS_NAME	RED	GREEN	BLUE	COUNT
0	0	1	RICE	255	255	0	1672064
1	1	2	BUILTUP	255	0	0	381868
2	2	3	WATERBODIES	0	112	255	618825
3	3	4	FL_BL	183	207	132	3265495
4	4	5	VEG_OC	230	0	169	8180422
5	5	6	SANDBAR	204	204	204	247390

Figure 4.9 Output: Count of pixels in each class

Now the crop area is calculated using field calculator in Attribute table using the following concept:

Area (in m) = Number of pixels * Area of each Pixel

where, Area of each pixel for Landsat 8 = 30 X 30 m

Note: Area in Hectare = Area in square meters / 10000

Table 4.4 Output: Area of each Class in Attribute Table

Table

LULC4

Rowid	VALUE	CLASSVALUE	CLASS_NAME	RED	GREEN	BLUE	COUNT	AREA_hec
0	0	1	RICE	255	255	0	1672064	150485.8
1	1	2	BUILTUP	255	0	0	381868	34368.12
2	2	3	WATERBODIES	0	112	255	618825	55694.25
3	3	4	FL_BL	183	207	132	3265495	293894.6
4	4	5	VEG_OC	230	0	169	8180422	736238
5	5	6	SANDBAR	204	204	204	247390	22265.1

Rice Crop Area (observed) = **150485.8 Hectare**

4.4.1 Accuracy Assessment - Crop Area Verification

$$Pa = 100 - \frac{\sqrt{(actual - observed)^2}}{\text{actual}} X\ 100$$

Where, Pa is percentage accuracy

Observed Rice Crop Area = **150485.8 Hectare**

Actual Rice Crop Area = **166544 Hectare**

Pa = 90.35 %

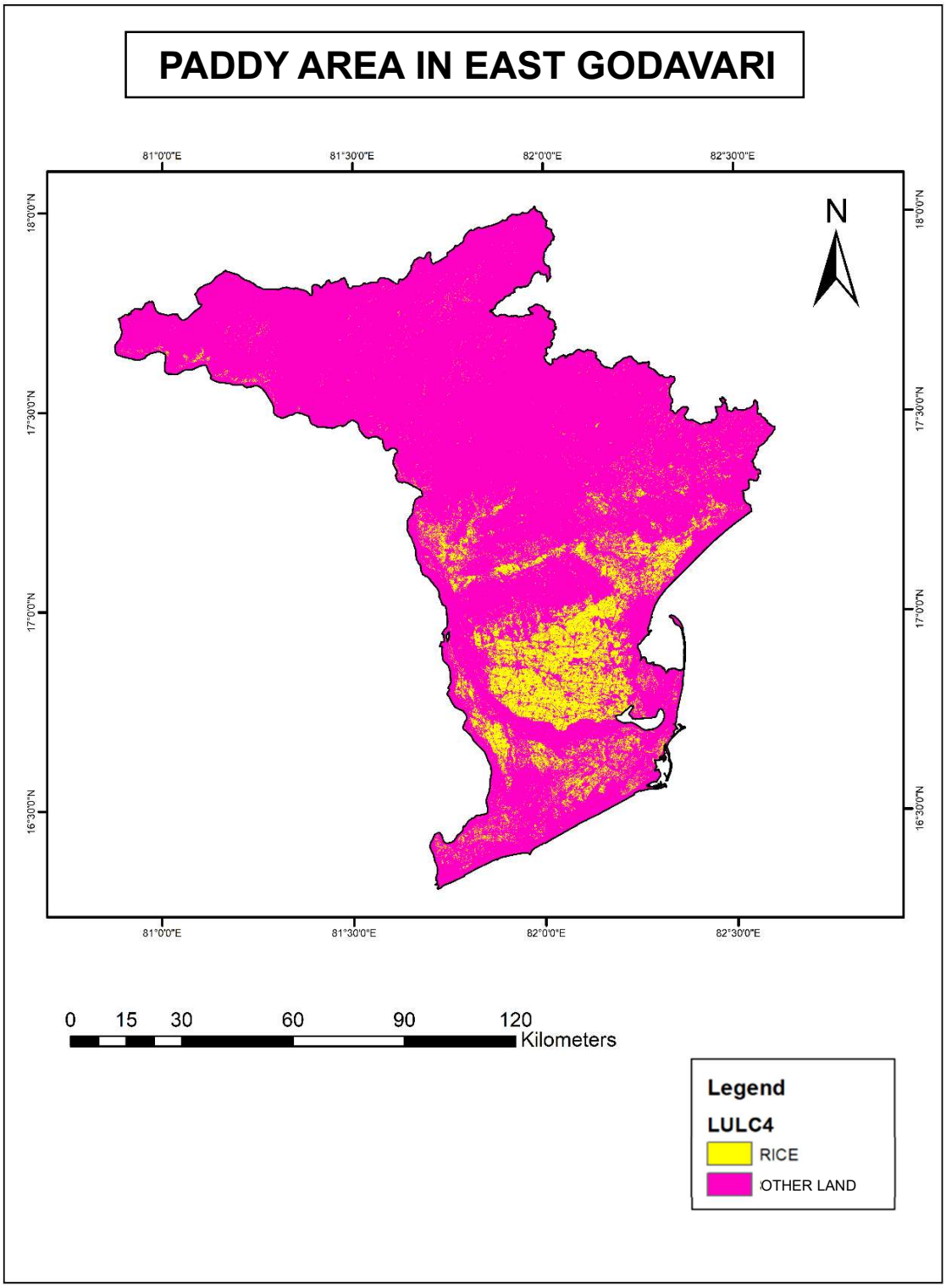


Figure 4.10 Map showing extent of Paddy in East Godavari

4.5 MAPPING THE NDVI OF CROP AREA

4.5.1 NDVI Estimation

Tool Used: ***Raster calculator***

The Raster Calculator tool allows you to create and execute a Map Algebra expression that will output a raster. Use the rasters list to select the datasets and variables to use in the expression. The Tools list provides a selection of commonly used conditional and mathematical tools, allowing you to easily add them to the expression. Numerical values (and mathematical operator symbols) can be added to the expression directly. Full paths to data or data existing in the specified current workspace environment setting can be entered in quotes (""). Numbers and scalars can be directly entered into an expression.

Spatial Analyst Tools ⟶ Map Algebra ⟶ **Raster Calculator**

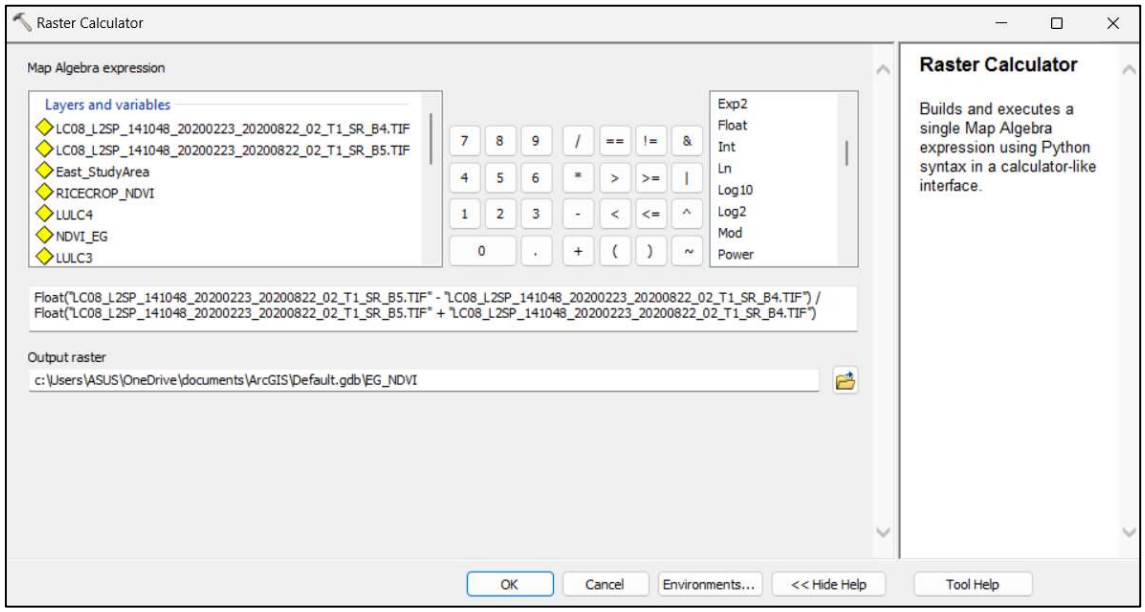


Figure 4.11 Input for NDVI

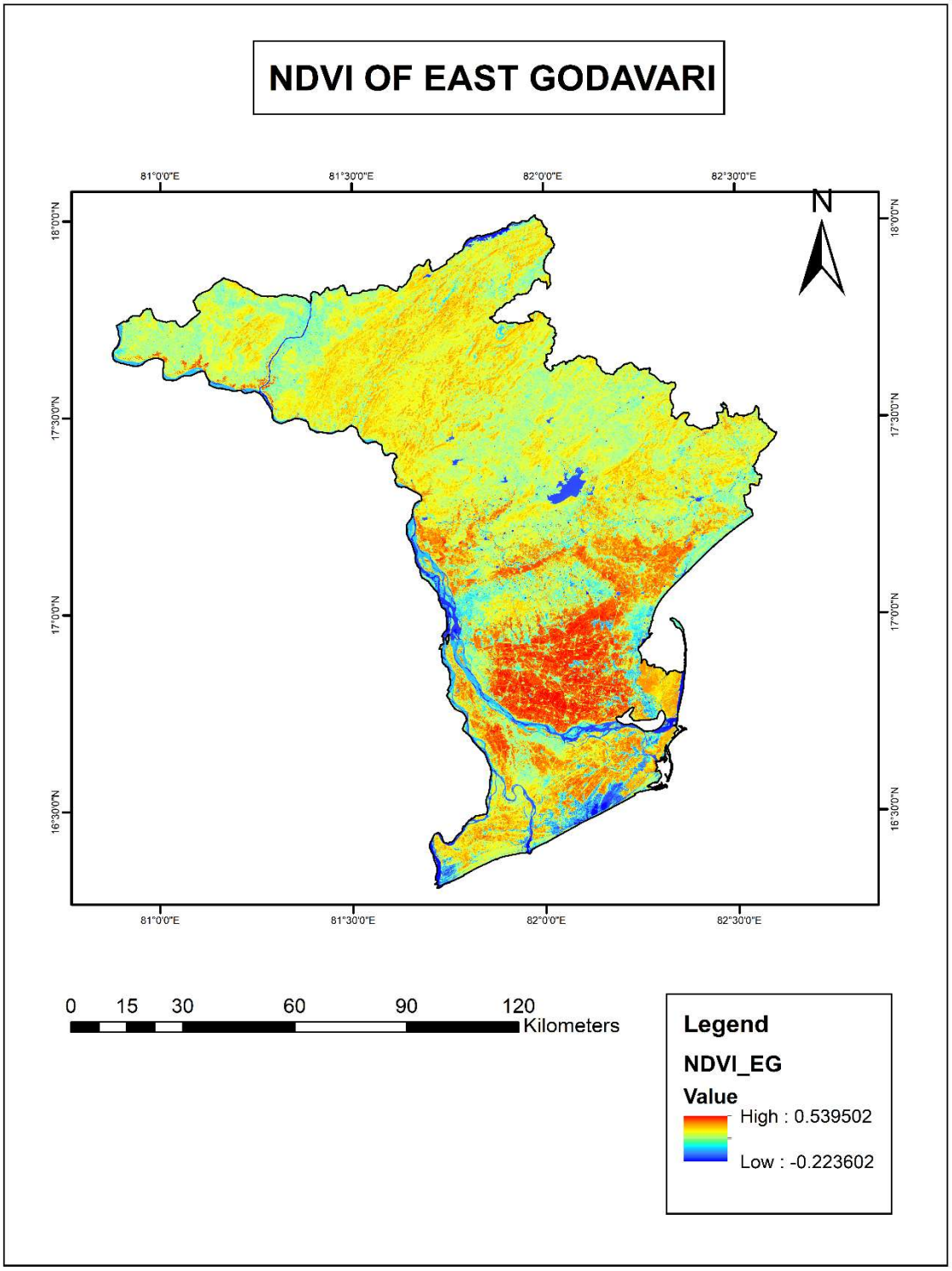


Figure 4.12 Output: NDVI map of East Godavari

4.5.2 Extraction of Crop Area NDVI

Tool Used: ***Con***

- It performs a conditional if/else evaluation on each of the input cells of an input raster.
- If either the true raster or optional false raster is floating point, the output raster will be floating point. If both the true expression and optional false raster are integer, the output raster will be integer. If the evaluation of the expression is nonzero, it is treated as True.
- If no input false raster or constant is specified, No Data will be assigned to those cells that do not result in True from the expression.
- If No Data does not satisfy the expression, it does not receive the value of the input false raster; it remains no Data.
- The Expression uses an SQL query. See the following topics for more details on creating queries in the Query Builder:

  Building a query expression and SQL reference for query expressions used in ArcGIS

Spatial Analyst Tools ⟶ Conditional ⟶ **Con**

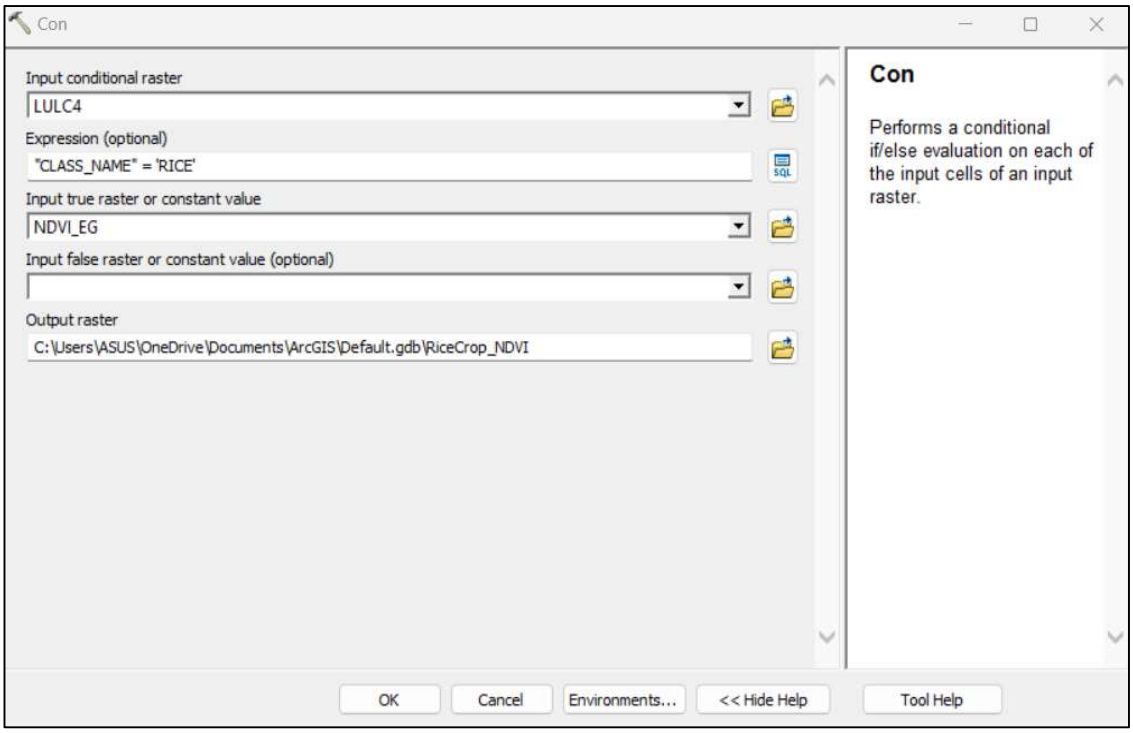


Figure 4.13 Input for extraction of Crop NDVI

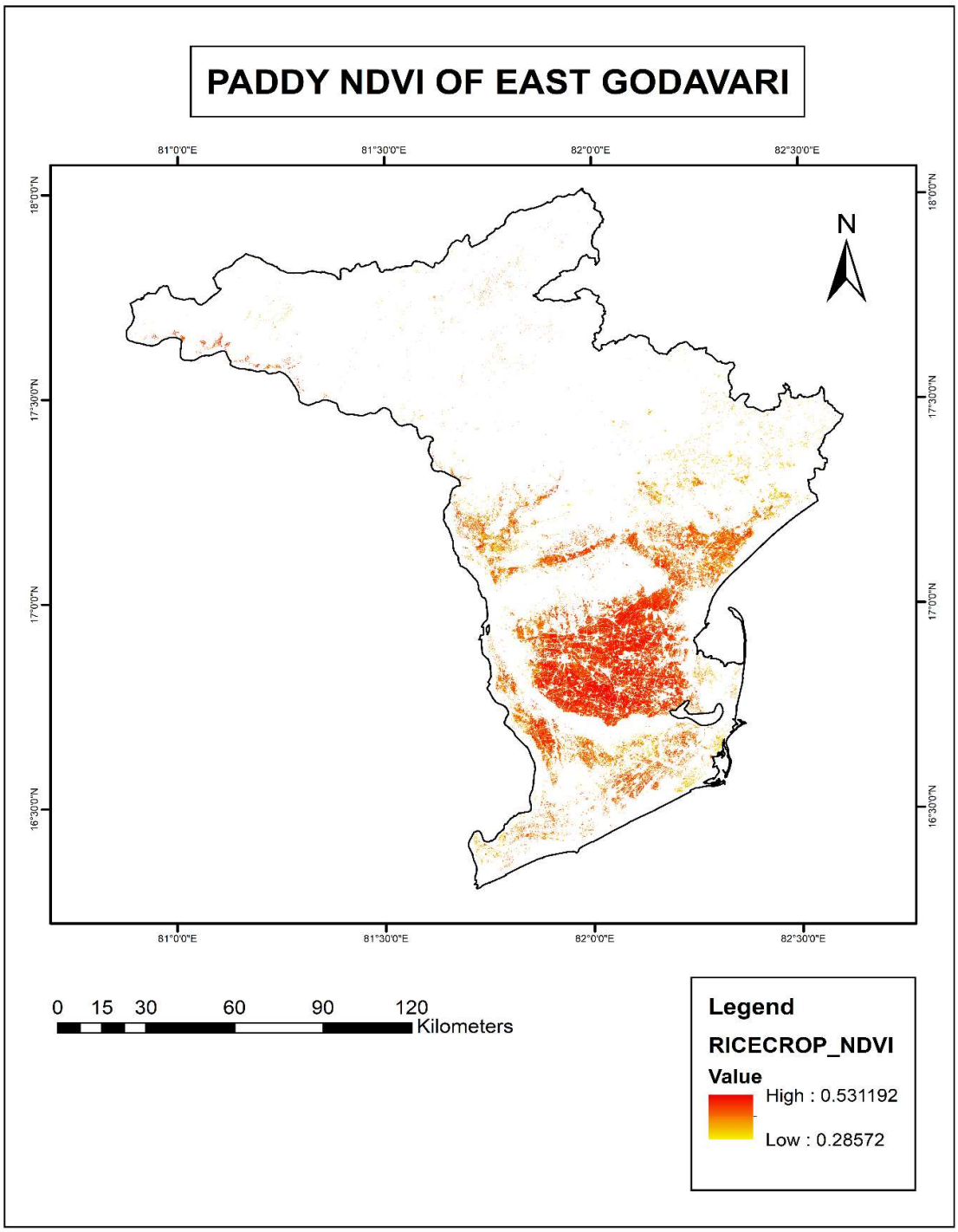


Figure 4.14 Output: NDVI of Paddy in East Godavari

Max NDVI = 0.531192

Min NDVI = 0.28572

4.6 ANALYSIS OF CROP YIELD AND NDVI

4.6.1 Relating Crop yield with NDVI

In this step, the Average Rice Crop NDVI value is related to the crop yield of study area (East Godavari) and then the maximum crop yield and minimum crop yield are calculated by interpolation.

Table 4.5 Crop Yield Estimation and interpolation

Crop NDVI	NDVI	Crop Yield (in Kg/Hectare)	Crop Yield Source
Max NDVI	0.531192	**6936.216**	Obtained by Interpolation
Min NDVI	0.28572	**3730.887**	Obtained by Interpolation
Average NDVI	**0.408456**	**5333.551**	**(From Table 4.1 crop data)**

4.6.2 Crop yield Equation Development

In this step, the above crop statistics are plotted as an exponential curve in Ms - Excel such that NDVI values are plotted on X – axis and crop yield values are plotted on Y – axis.
Based on the exponential curve Ms- Excel is capable to generate the exponential equation for Crop Yield in terms of NDVI. The output or results are shown below:

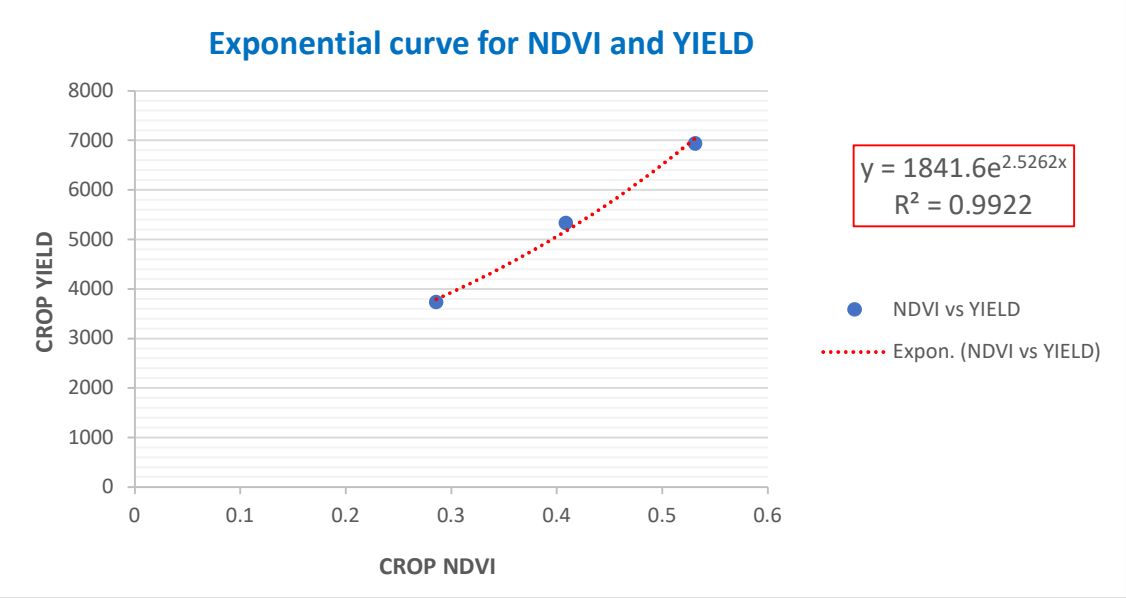


Figure 4.15 Output: Exponential Curve for NDVI & Crop Yield and Crop Yield Equation

From the above chart, the yield equation is given by

$$y = 1841.6e^{2.5262x}$$

where, a = 1841.6 b = 2.5262 x = Rice crop NDVI y = Yield

4.7 MODEL DEVELOPMENT IN ARCGIS

Model Builder is a graphical user interface (GUI) tool in ArcGIS that allows you to create, edit, and manage geoprocessing workflows. It provides a way to automate complex GIS tasks by connecting pre-built tools and scripts into a sequence of tasks called a model.

Note: Model parameters are user-defined inputs that allow users to modify the behaviour of a model created in Model Builder in ArcGIS.

Tool Used: ***Model builder***

It is used to make a geoprocessing model. We can access the model builder tool from Standard Toolbar as shown below:

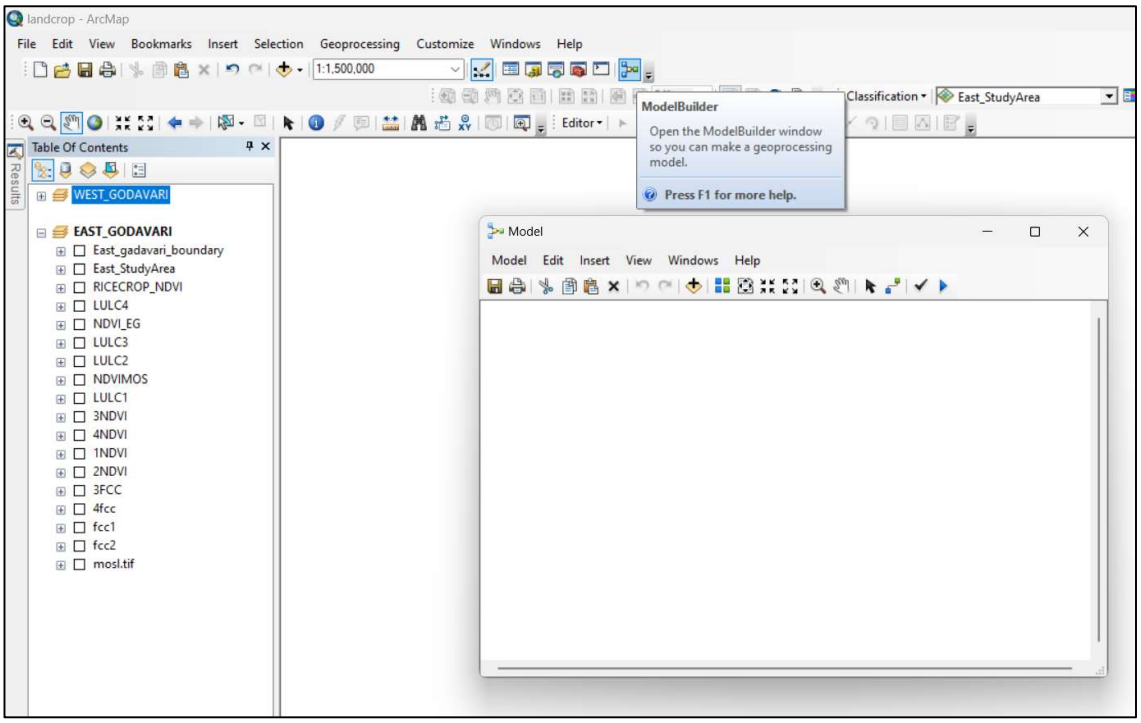


Figure 4.16 Model Builder Tool in Standard Toolbar of ArcMap

Now create a model as shown in below figure:

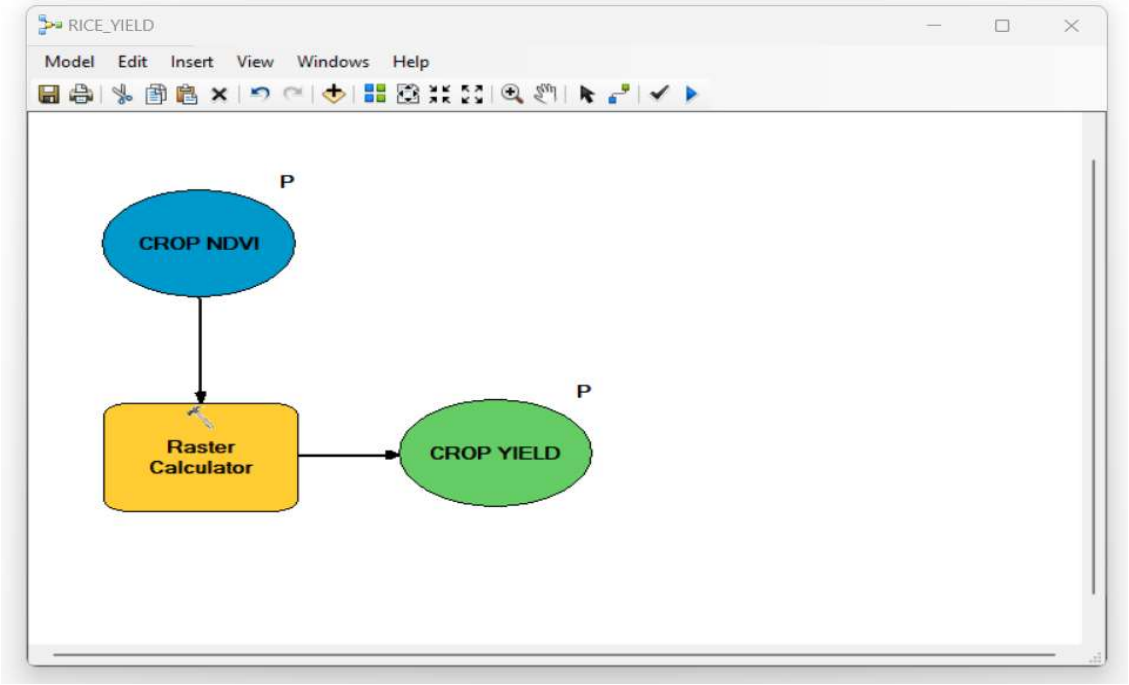


Figure 4.17 Crop Yield Model

In the above Figure, Crop NDVI entity acts as input parameter and Crop Yield entity acts as output parameter. Raster Calculator acts as a function that processes the input and gives the output based on the given expression (yield equation) as shown below:

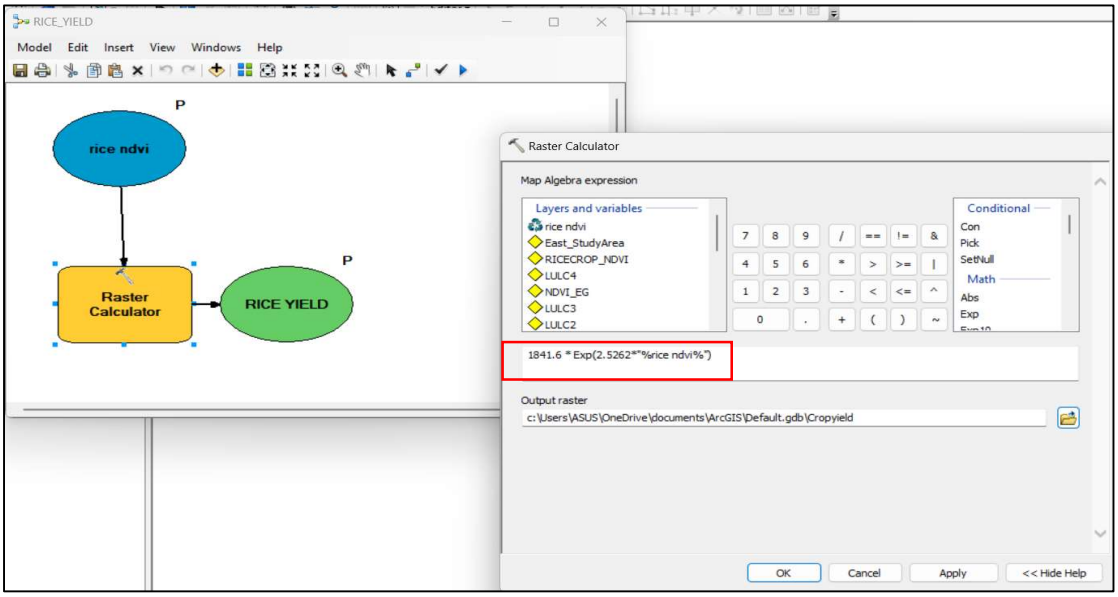


Figure 4.18 Input expression for Raster calculator

Now save the model in the output location with name “Rice_Yield”. It saves the model in (.tbx) which works as tool in ArcGIS.

Open Arc toolbox and import the saved “Rice_Yield.tbx” tool. Thus, Crop Yield tool (model) is created in ArcGIS.

4.8 CROP YIELD MAP GENERATION USING CROP YIELD MODEL

Tool Used: ***Rice_Yield***

Arc Toolbox ⟶ RICEMODEL ⟶ **RICE_YIELD**

Open Arc toolbox then go to Rice_Yield tool and double click on it and a new window appears then give the Rice Crop NDVI of East Godavari as Input file and specify the output location and name.

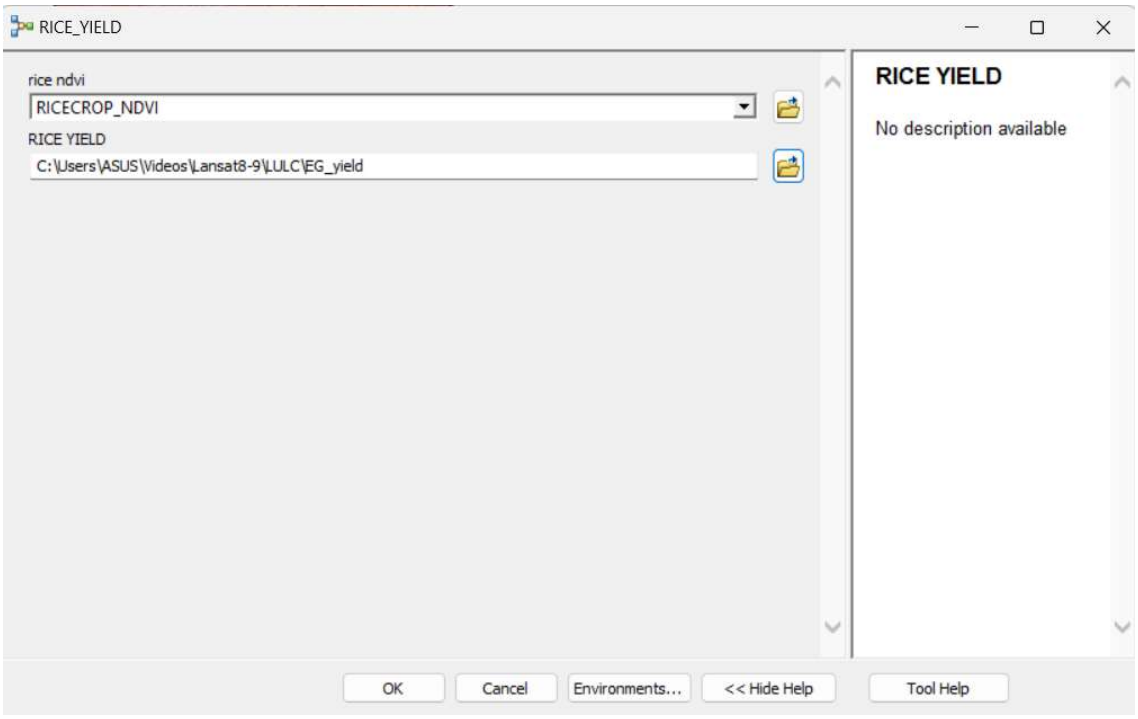


Figure 4.19 Input for Crop Yield Map

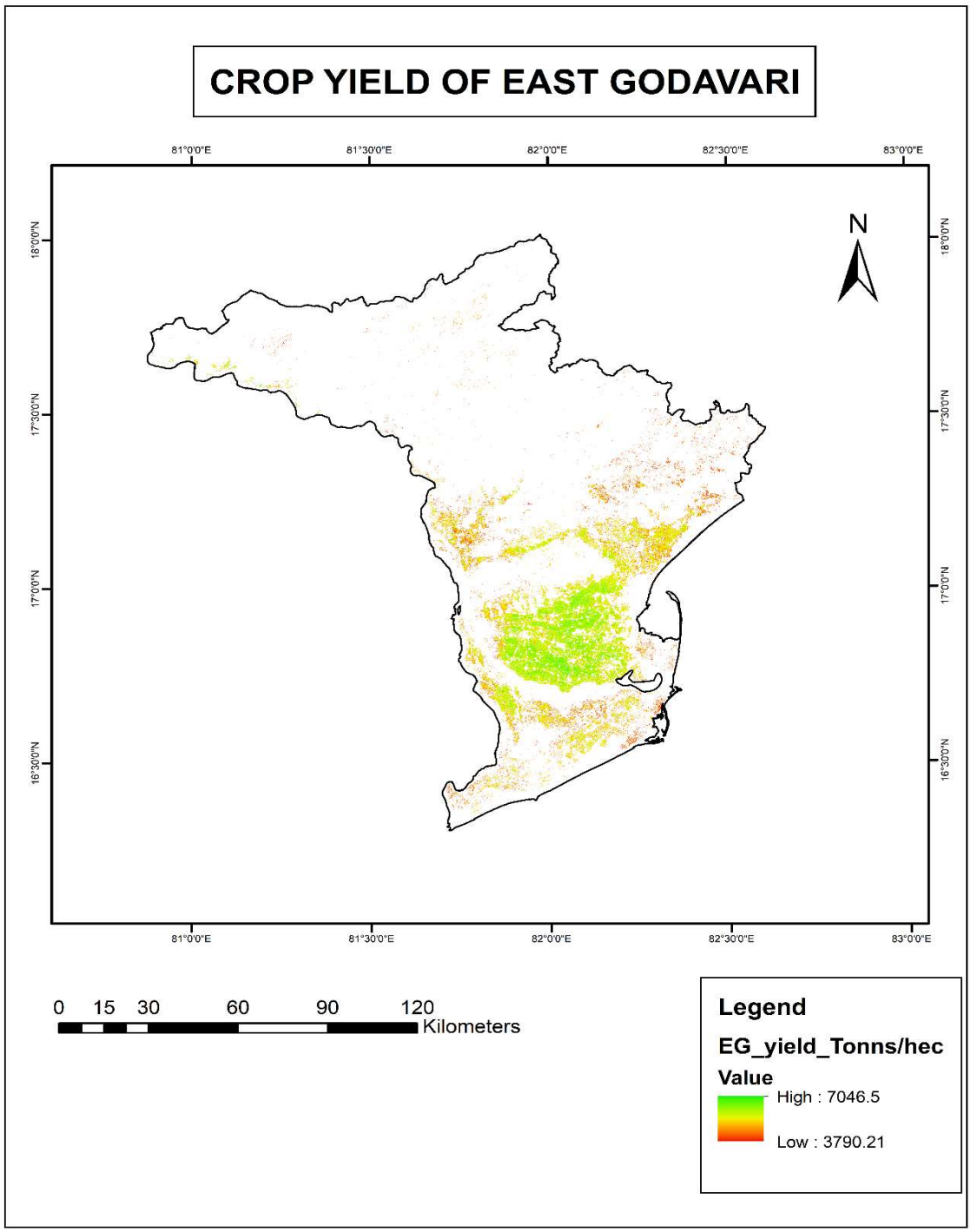


Figure 4.20 Output: Paddy Yield Map of East Godavari

Max yield = 7046.5 Kg/hectare

Min yield = 3790.21 Kg/hectare

4.9 ACCURACY ASSESSMENT - CROP YIELD VERIFICATION

Table 4.6 Estimated Crop Yield Observations in East Godavari

Crop yield	Crop Yield (in Kg/hectare)	Actual Crop Yield
Max yield	7046.50	-
Min yield	3790.21	-
Average yield	**5418.36**	**5333.551**

$$Pa = 100 - \frac{\sqrt{(actual - observed)^2}}{\text{actual}} X\ 100$$

Where, Pa is percentage accuracy

Observed Rice Crop Yield = **5418.36 Kg/Hectare**

Actual Rice Crop Yield = **5333.551 Kg/Hectare**

Pa = 98.41 %

Now we are going to validate this model on West Godavari District.

4.10 MODEL VALIDATION ON ANOTHER STUDY AREA

4.10.1 FCC Generation of West Godavari:

FCC (*False Color Composite*) is useful for analyzing vegetation. In particular, areas in red have better vegetation health. Dark areas are water and urban areas are white.
It uses near-infrared (Band 5), red (Band 4), and green (Band 3). Because chlorophyll reflects near-infrared light.

Tool Used: ***composite Bands***

Data Management Tools ⟶ Raster ⟶ Raster Processing ⟶ **Composite Bands**

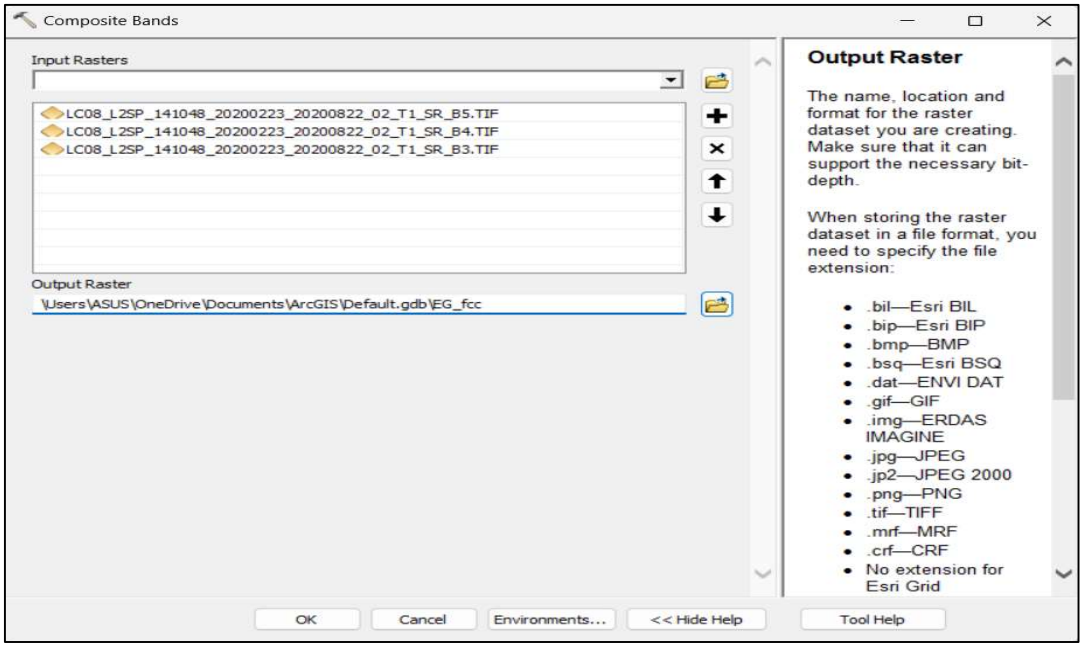


Figure 4.21 Input for FCC generation

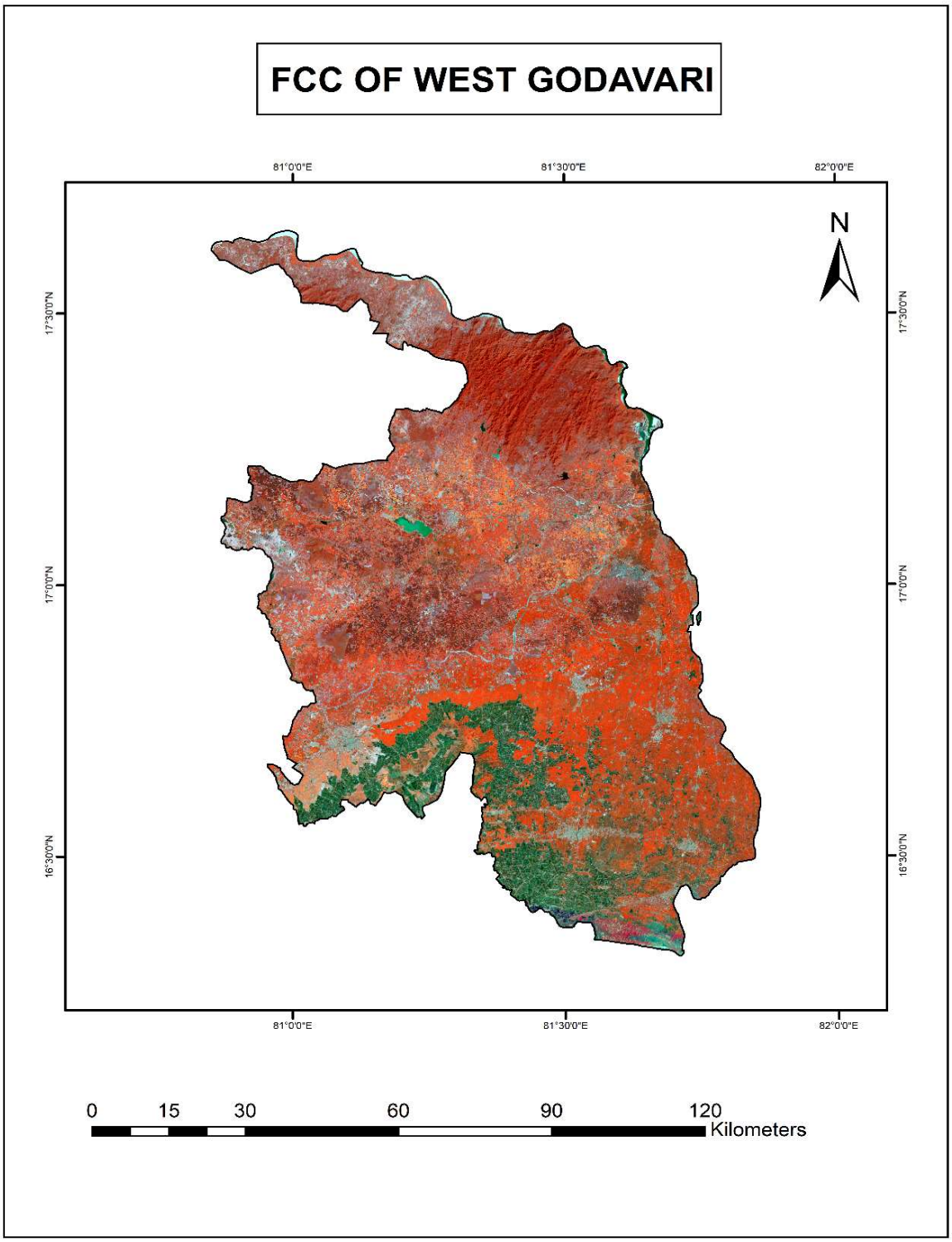


Figure 4.22 Output: FCC map of West Godavari

4.10.2 Crop classification of West Godavari using SVM (support Vector Machine):

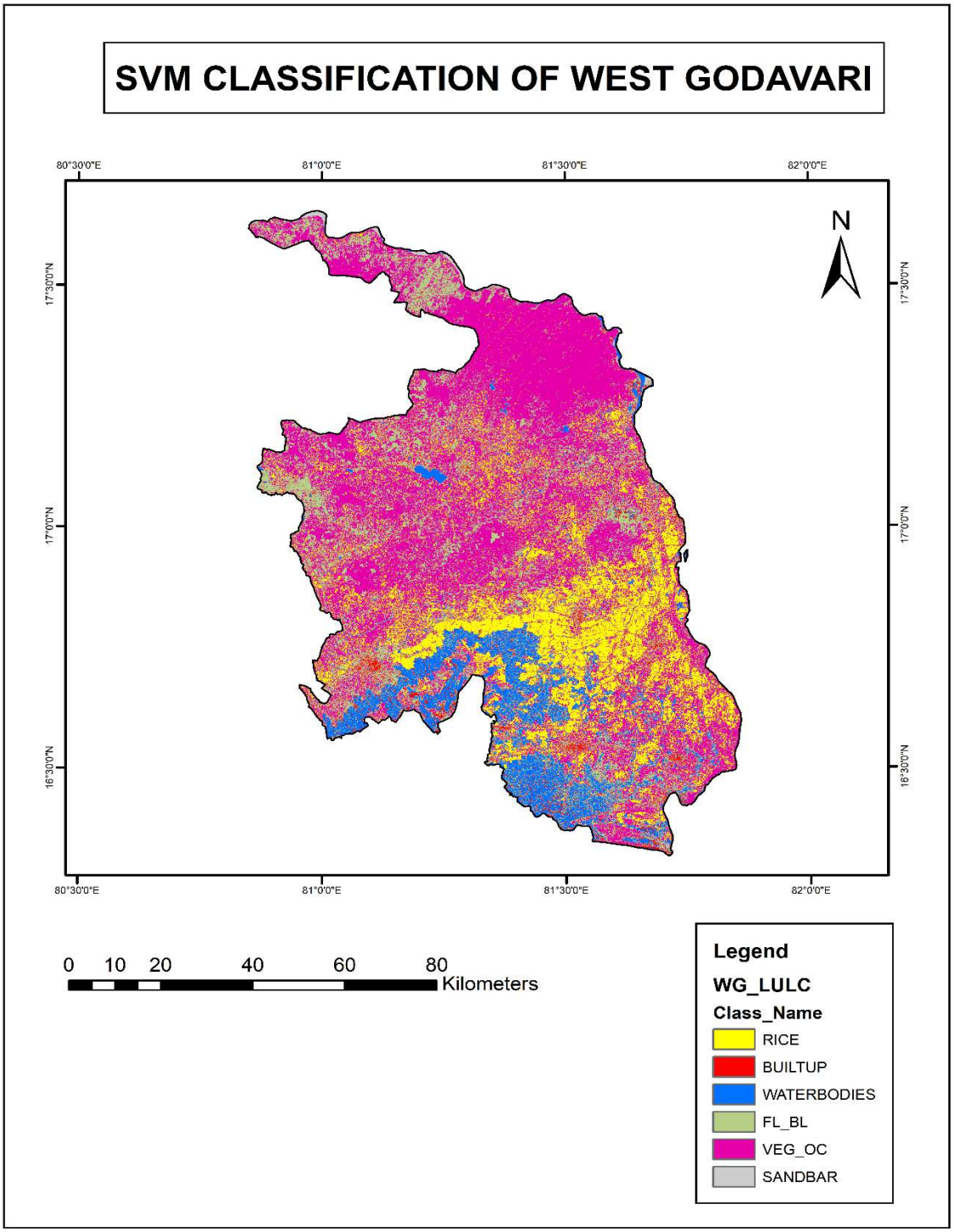


Figure 4.23 Output: SVM classified map of West Godavari

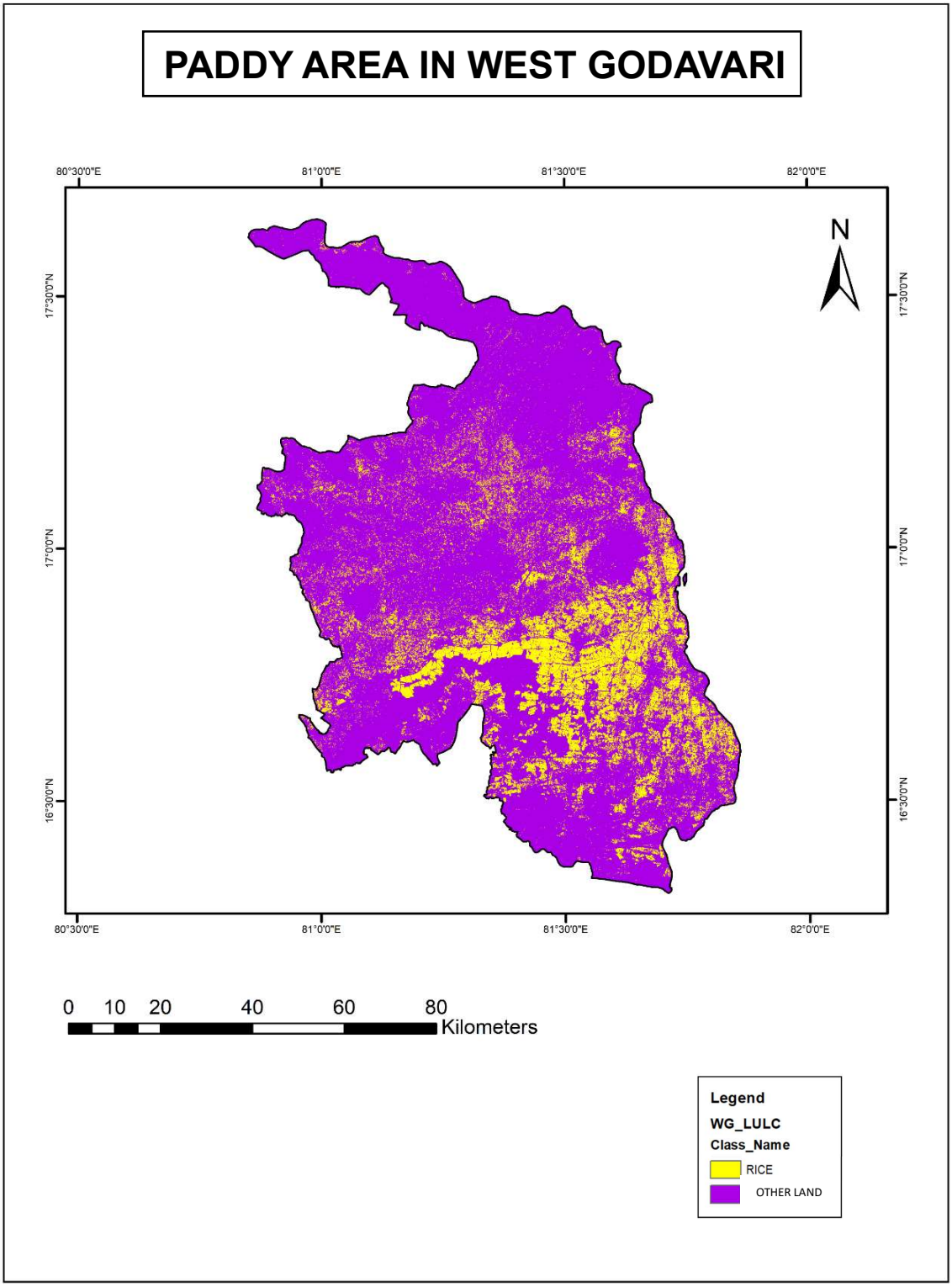


Figure 4.24 Map showing extent of Paddy in West Godavari

4.10.3 NDVI of West Godavari:

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Tool Used: **Raster calculator**

The Raster Calculator tool allows you to create and execute a Map Algebra expression that will output a raster. Use the rasters list to select the datasets and variables to use in the expression. The Tools list provides a selection of commonly used conditional and mathematical tools, allowing you to easily add them to the expression. Numerical values (and mathematical operator symbols) can be added to the expression directly. Full paths to data or data existing in the specified current workspace environment setting can be entered in quotes (""). Numbers and scalars can be directly entered into an expression.

Spatial Analyst Tools ⟶ Map Algebra ⟶ **Raster Calculator**

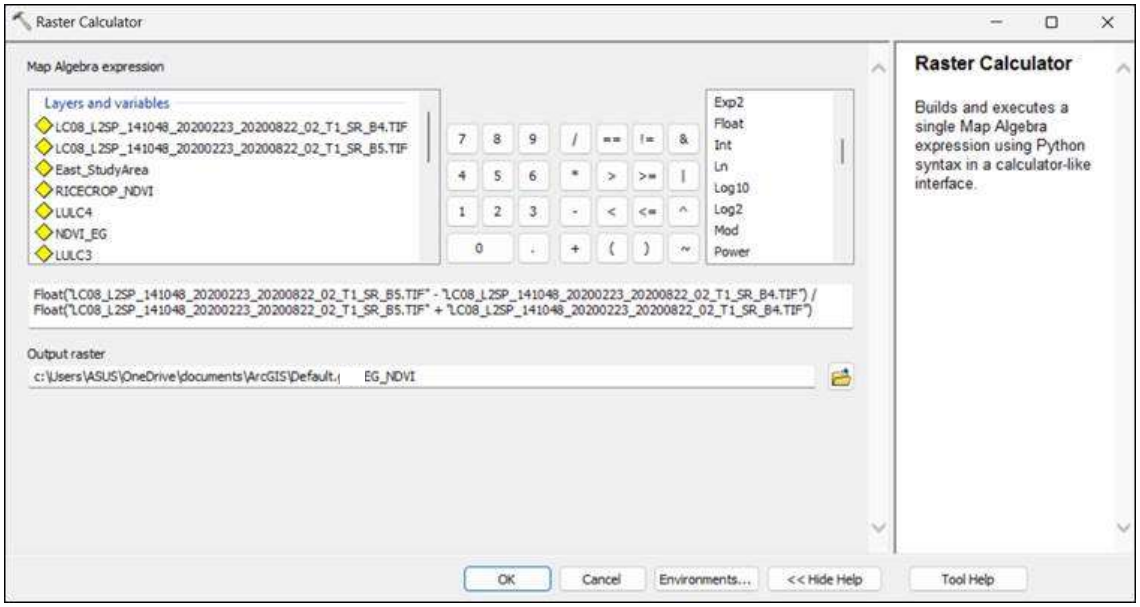


Figure 4.25 Input for NDVI

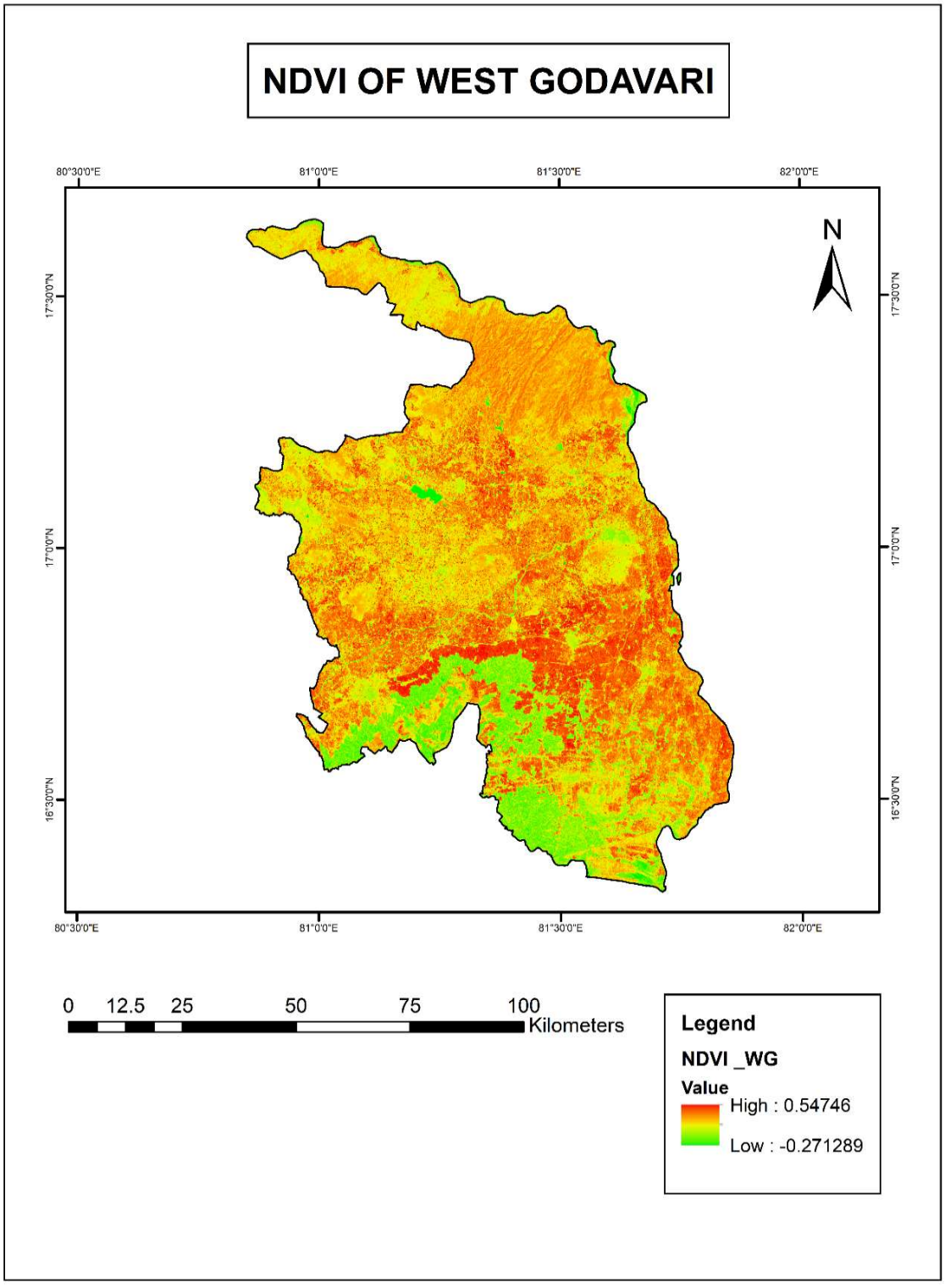


Figure 4.26 NDVI Map of West Godavari

4.10.4 Extraction of Crop Area NDVI in West Godavari

Tool Used: ***Con***

- It performs a conditional if/else evaluation on each of the input cells of an input raster.
- If either the true raster or optional false raster is floating point, the output raster will be floating point. If both the true expression and optional false raster are integer, the output raster will be integer. If the evaluation of the expression is nonzero, it is treated as True.
- If no input false raster or constant is specified, No Data will be assigned to those cells that do not result in True from the expression.
- If No Data does not satisfy the expression, it does not receive the value of the input false raster; it remains no Data.
- The Expression uses an SQL query. See the following topics for more details on creating queries in the Query Builder:
  Building a query expression and SQL reference for query expressions used in ArcGIS

Spatial Analyst Tools ⟶ Conditional ⟶ **Con**

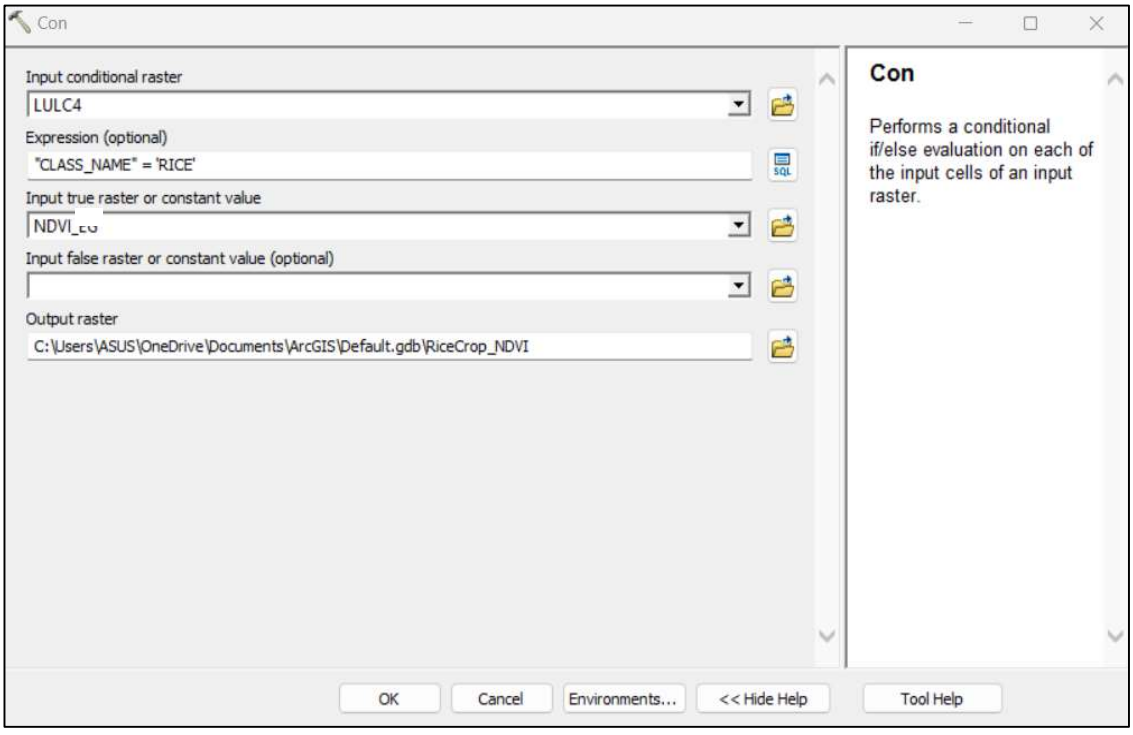


Figure 4.27 Input for extraction of Crop NDVI

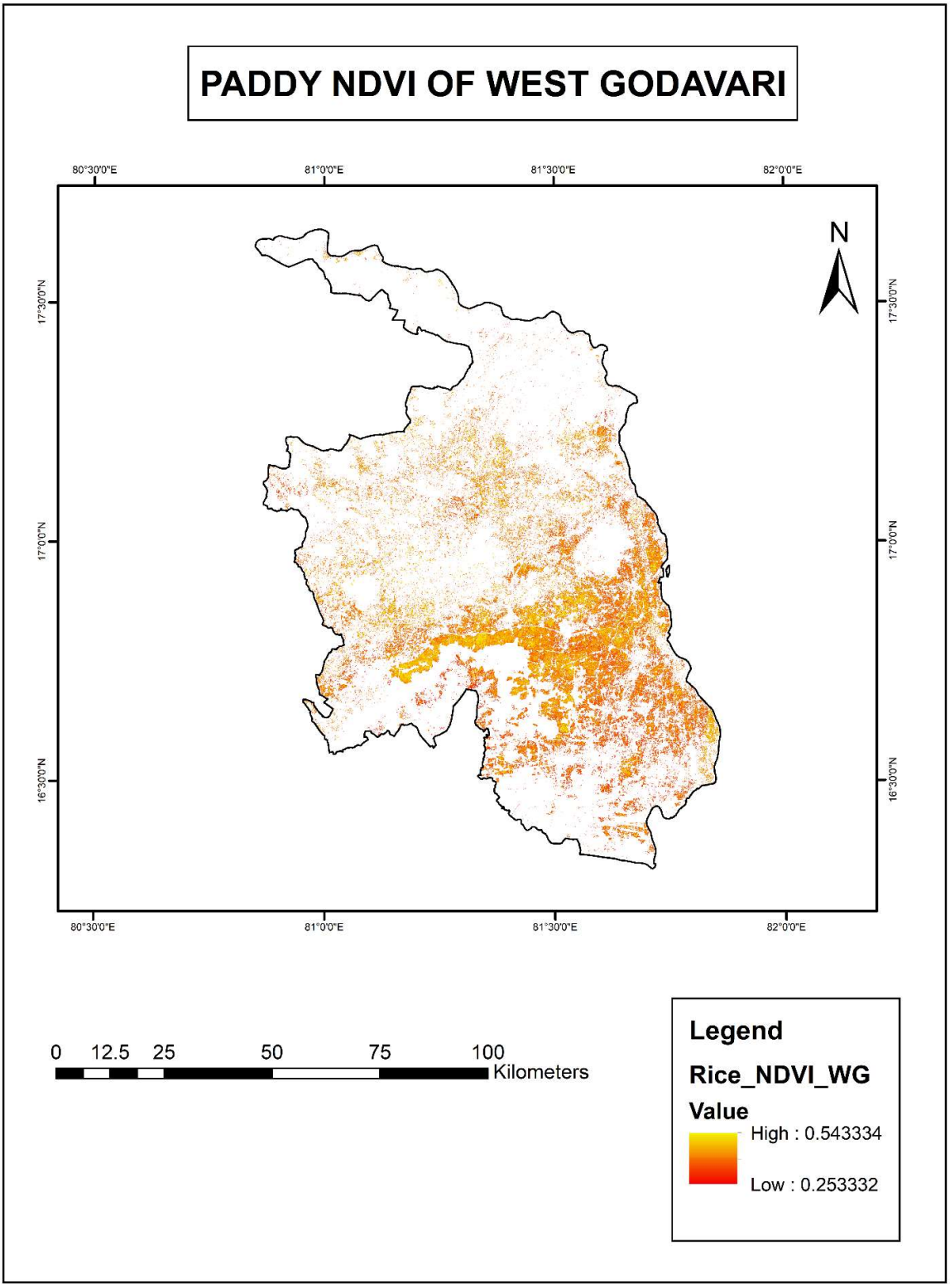


Figure 4.28 NDVI of Paddy in West Godavari

4.10.5 Crop Yield Map Generation Using Crop_Yield Model in West Godavari

Tool Used: ***Rice_Yield***

Arc Toolbox ⟶ RICEMODEL ⟶ **RICE_YIELD**

Open Arc toolbox then go to *Rice_Yield tool* and double click on it and a new window appears then give the *Rice Crop NDVI of West Godavari* as Input file and specify the output location and name.

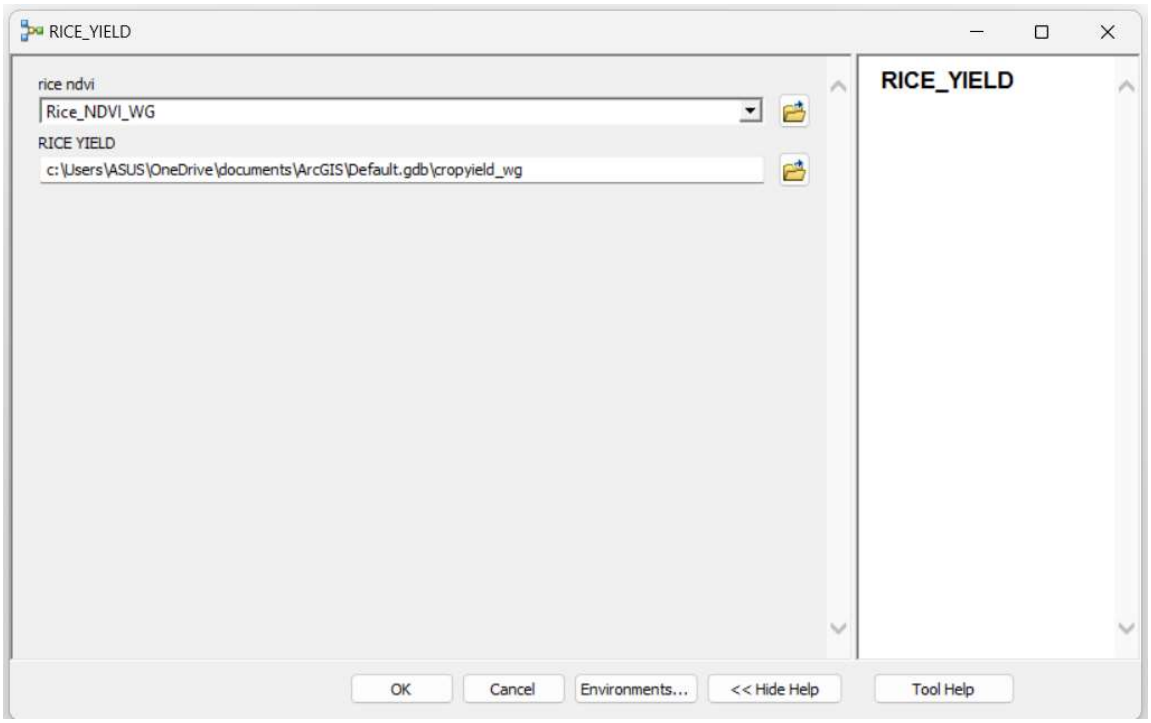


Figure 4.29 Input for Crop Yield Map of West Godavari

The Outputs are discussed in the Results Chapter.

CHAPTER - 5

RESULTS & DISCUSSION

5.1 PADDY YIELD MAP OF WEST GODAVARI

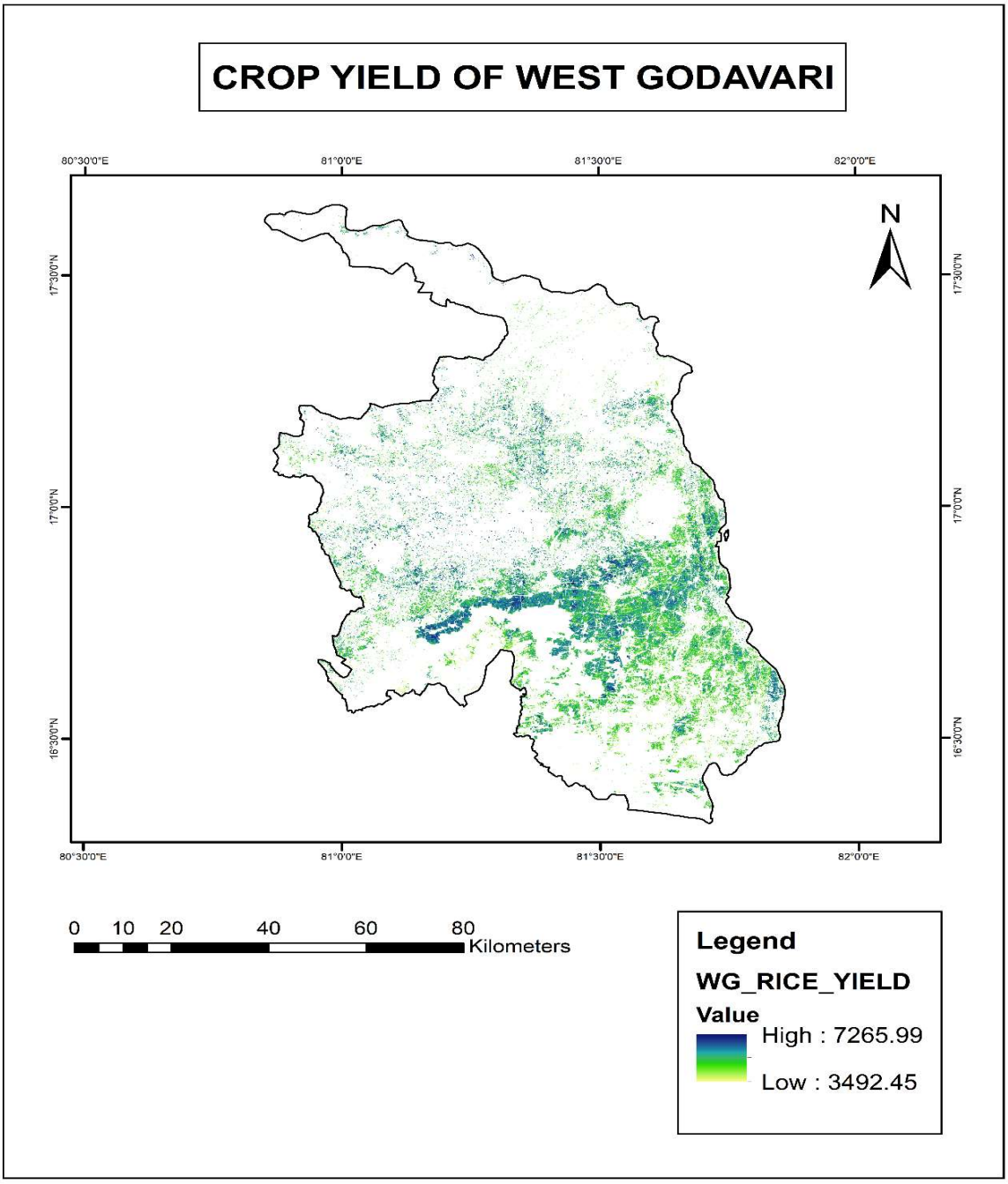


Figure 5.1 Output: Paddy Yield Map of West Godavari

5.2 CROP YIELD VERIFICATION IN WEST GODAVARI

Table 5.1 Estimated Crop Yield Observations in West Godavari

Crop yield	Crop Yield (in Kg/hectare)	Actual Crop Yield (in Kg/hectare)
Max yield	7265.99	-
Min yield	3492.45	-
Average yield	**5379.22**	**5394.66**

$$Pa = 100 - \frac{\sqrt{(actual - observed)^2}}{\text{actual}} X\ 100$$

Where, Pa is percentage accuracy

Observed Rice Crop Yield = **5379.22 Kg/Hectare**

Actual Rice Crop Yield = **5394.66 Kg/Hectare**

Pa = 99.71 %

5.3 EVALUATION OF OUTCOMES

Through integration of remotely sensed data, an assessment of field level yield prediction has been executed. This study established that there is a significant positive relationship between remotely sensed Normalised Difference Vegetation Index and field level yield, where production is at the mercy of many factors acting on the crop. This clearly shows the potential of using NDVI for yield prediction at field level.

Figure 4.15 shows that the crop yield equation indicating the relationship between Crop yield and NDVI index. These are highly correlated as $R^2 = 0.9922$. and Figure 4.20 shows that the crop yield estimation in East Godavari that is generated based on the equation and the result is 98 percent accurate. Again, the estimated crop yield in West Godavari resulted 99 percent accurate as shown in figure 5.1.

CHAPTER - 6
CONCLUSIONS

Traditionally, crop yield estimation has been done mainly through crop surveys, and this process can be labour intensive and time consuming. This can lead to slower yield estimation because the data takes considerable time to collect. As technology has advanced, satellite-based remote sensing has become more popular for monitoring and observing vegetation growth and estimation of crop yield. There are a number of advantages regarding remote sensing for crop yield estimation such as lower cost, faster in obtaining and updating data, and data from remote sensing is also relatively accurate.

The purpose of this study was to propose a method to estimate the crop yield of an area with the help of single date multi-spectral image for yield prediction and to establish the relationship between NDVI and field level crop yield. In this study, a simple but holistic methodology was developed to identify the relationship between NDVI and Crop Yield and to develop an ArcGIS tool which can easily generates the Crop yield map with the help of single input of Crop NDVI.

The results are reliable about 90 percent accurate using SVM classification for classifying the crops in the study area. And the yield equation representing the relation between NDVI and crop yield is highly correlated as the coefficient of determination (R^2) value is 0.9922. Finally, the estimated crop yield using the yield equation has an accuracy of 98 percent.

In conclusion, The Normalized Difference Vegetation Index (NDVI) is a widely used remote sensing index for monitoring crop growth and yield estimation. This study has shown that Landsat - NDVI can be used effectively to predict crop yields across the district one to two months before harvest. In general, the predicted yields were ±10% of the actual observed yields. Although the developed models show some promise in crop yield forecasting on the East Godavari and West Godavari, it has to be noted that all models have limitations. The NDVI measures potential yield; therefore, anything that happens to the crop after the forecast date is not reflected in the crop yield estimate. For example, if a drought, disease or pests outbreak happens after the forecast date the models would most likely overestimate crop yield. In addition, satellite images are affected by various atmospheric effects e.g., clouds and volcano eruptions, which in turn reduce the quality of the data acquired and subsequently the developed crop-yield models. Owing to the limited number of years used in the current study, the developed models have to be

constantly updated as both Landsat -NDVI and crop yield data become available. In addition, it has been shown that in the research of spring wheat yields on the Canadian Prairies can be forecast from indicators related to water stress *(Qian et al., 2009)*, and water demand and water balance *(Mkhabela et al., 2010)* at different crop stages. Meanwhile, *Boken and Shaykewich (2002)* combined a wheat yield model with NDVI to improve wheat yield predictions in Saskatchewan. *Vicente-Serrano et al. (2006)* combined NDVI and SPI, *while Rojas (2007)* combined the FAO crop specific water balance model with NDVI and meteorological data to improve crop yield forecasting in Spain and Kenya, respectively. Similarly, *Prasad et al. (2006)* developed two models that combined NDVI, soil moisture, surface temperature and rainfall data to forecast maize and soybean yield in Iowa, USA, while *Balaghi et al. (2008)* developed models that combined NDVI, rainfall and air temperature to forecast wheat yield in Morocco. All these studies reported improvements in crop yield predictions. Therefore, there is the possibility of combining NDVI data with climatological data, such as, the Standardised Precipitation Index (SPI) and evapotranspiration (ET) to improve the performance of the models, which is an obvious next step to be explored.

6.1 Future scope of work:

Future research could further explore the integration of land management techniques, climatological data with GOSAVI and GSAVI instead of NDVI. They use the green band instead of the red band, and so are somewhat different than the traditional NDVI, that uses the red band. RDVI was able to perform similarly to indices that use the green band, even though it uses the red band. RDVI is designed to provide balance between an environment with dense biomass and sparse vegetation, and maybe suitable for agricultural land. RDVI also has a relatively simple equation and relatively favourable performance.

There is need for further investigation on the other factors that contribute to yield variability at field level. There is need for detailed study on the soil characteristics at field level and include factors like pH, texture for specific areas / fields which have a significant effect on water holding capacity, nutrient availability and on yield.

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